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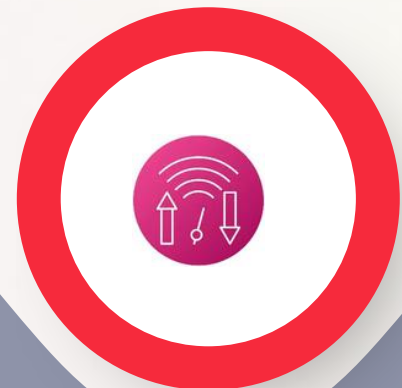
26th November, 2025

SILICON PHOTONIC RING MODULATORS

SILICON PHOTONIC RING MODULATORS

State of the Art -The Modulator Landscape

- High-speed data transmission:
- Modern optical interconnects.



Motivation - The Data Bandwidth Problem

- Data transmission
- Bandwidth demand in communication systems



Why Ring Modulators?

- Compact,
- Energy-efficient links
- Ultra-fast on-chip modulation.



Working Principle -Ring Physics

- Physics of silicon microring resonators
- Resonance condition,
- Thermo-optic vs carrier-based modulation, bandwidth



Simulation - Our Modelling Approach

- Methodology, models, and tools
- Device parameters from literature
- Electrical-optical co-design
- Performance metrics

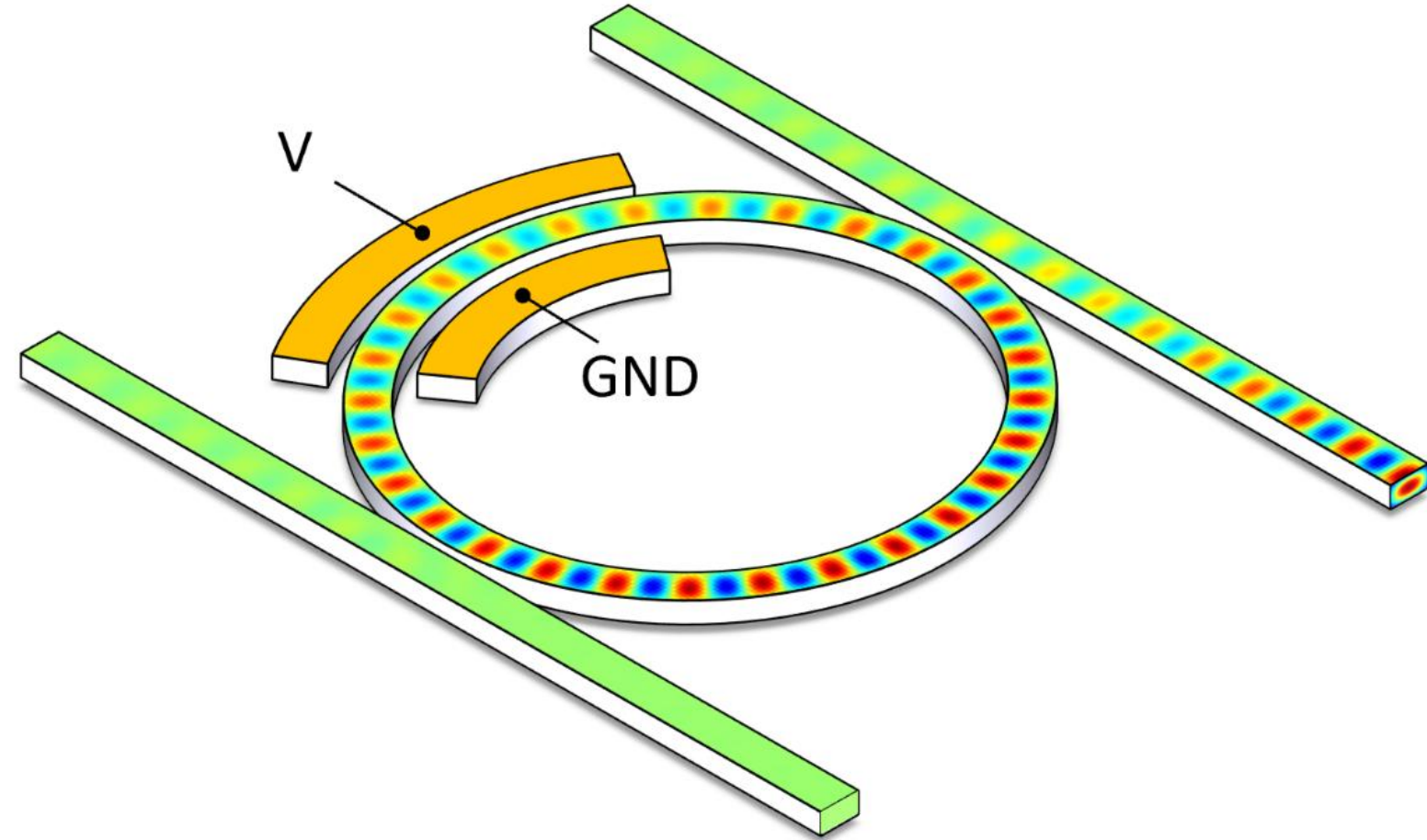
Results, Discussion & Future Possibilities

- Simulations results
- compare with reported 50 Gb/s and 100-200 Gb/s ring modulators.
- Limitations, design trade-offs
- Future possibilities for faster



AGENDA

- Motivation
- Working Principle
- State of the art
- Simulation
 - Results
 - Discussions
- Future possibilities
- Sources



MOTIVATION

MOTIVATION(1)

Electrical To Optical Interconnects

- Growing demands of internet
- Electrical Interconnect Bandwidth Bottleneck
- To ensure low energy consumption

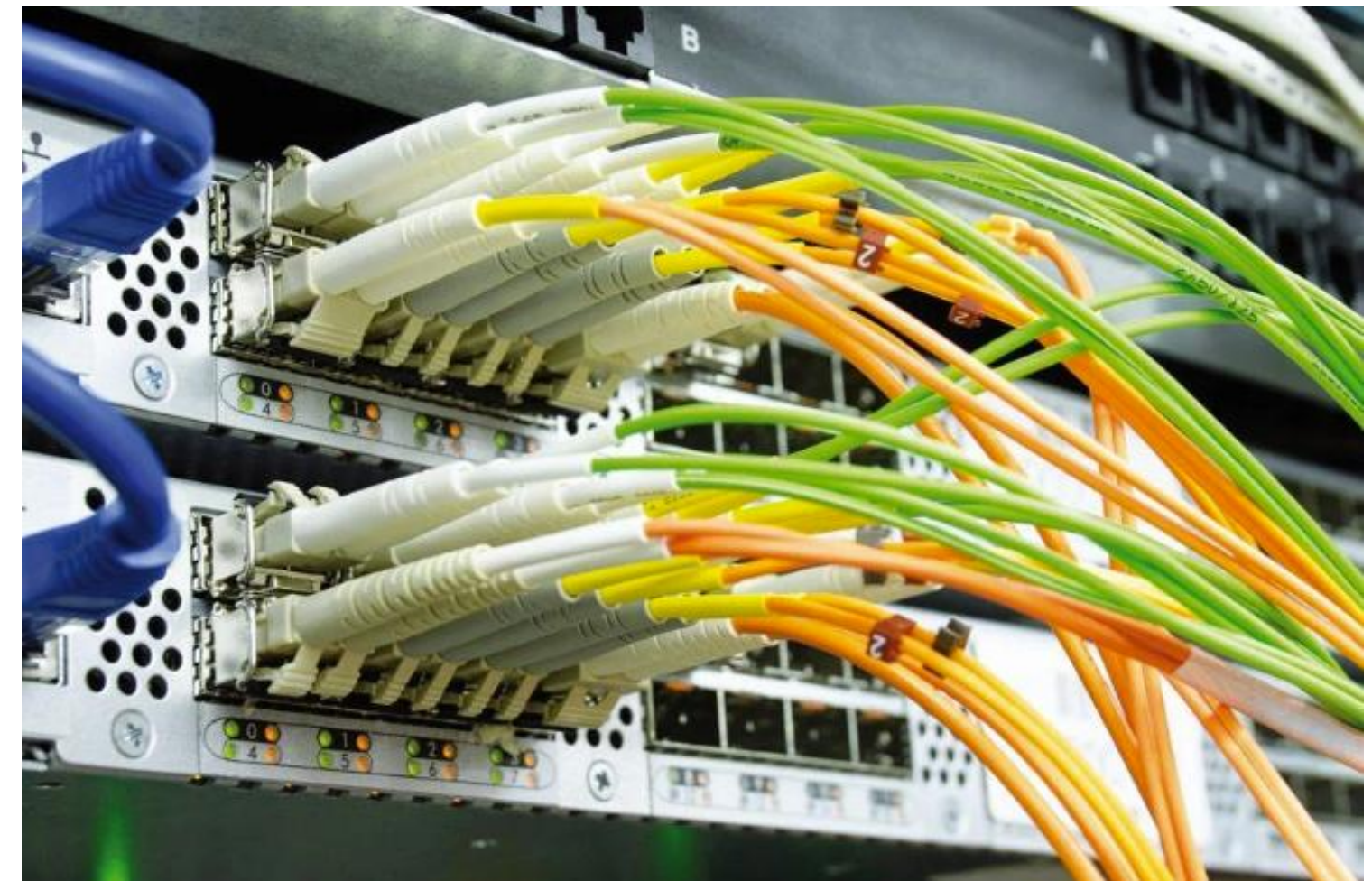


Image [source](#)

$$C = B \log_2(1 + SNR)$$

C: Channel capacity(bits/s)

B: Bandwidth(Hz)

SNR: Signal-to-noise ratio

$$EDP = E_{bit}t_{delay} \approx \frac{1}{2}CV^2.RC = \frac{1}{2}RC^2V^2$$

EDP : Energy Delay Product

E_{bit} : Energy per bit

t_{delay} : Delay per bit

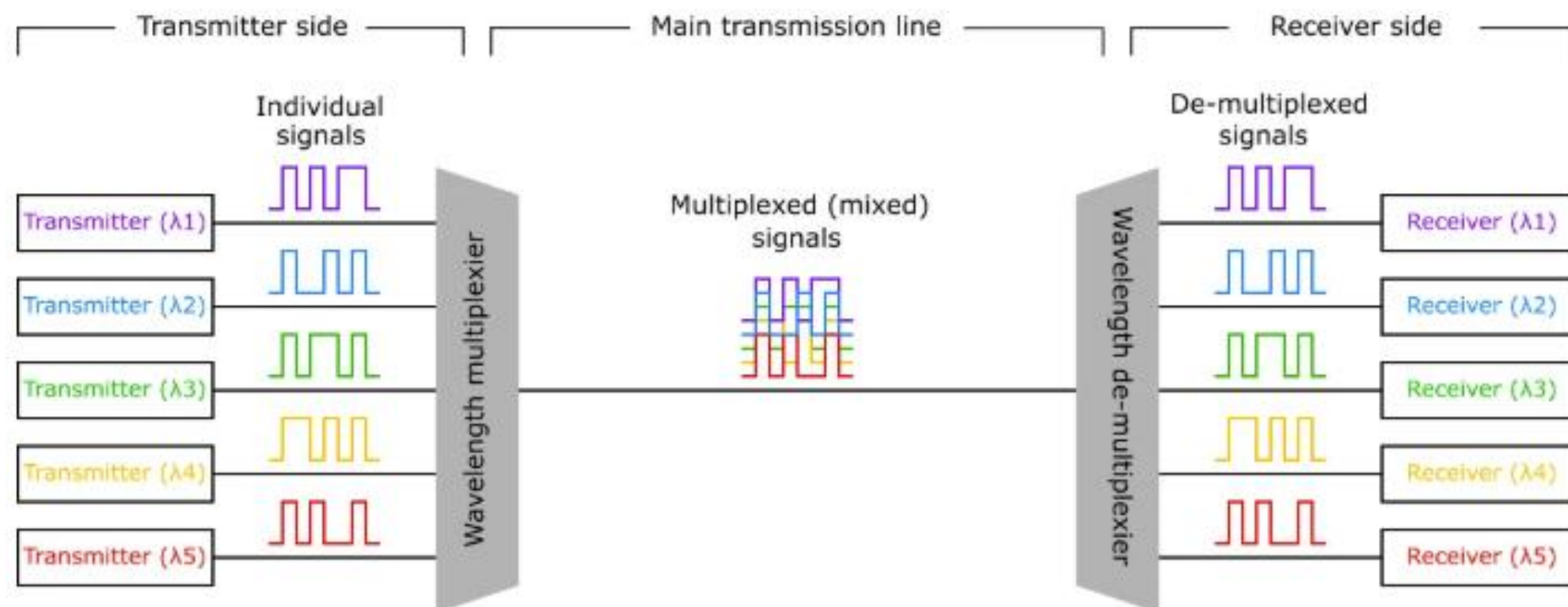
R : Resistance

C : Capacitance

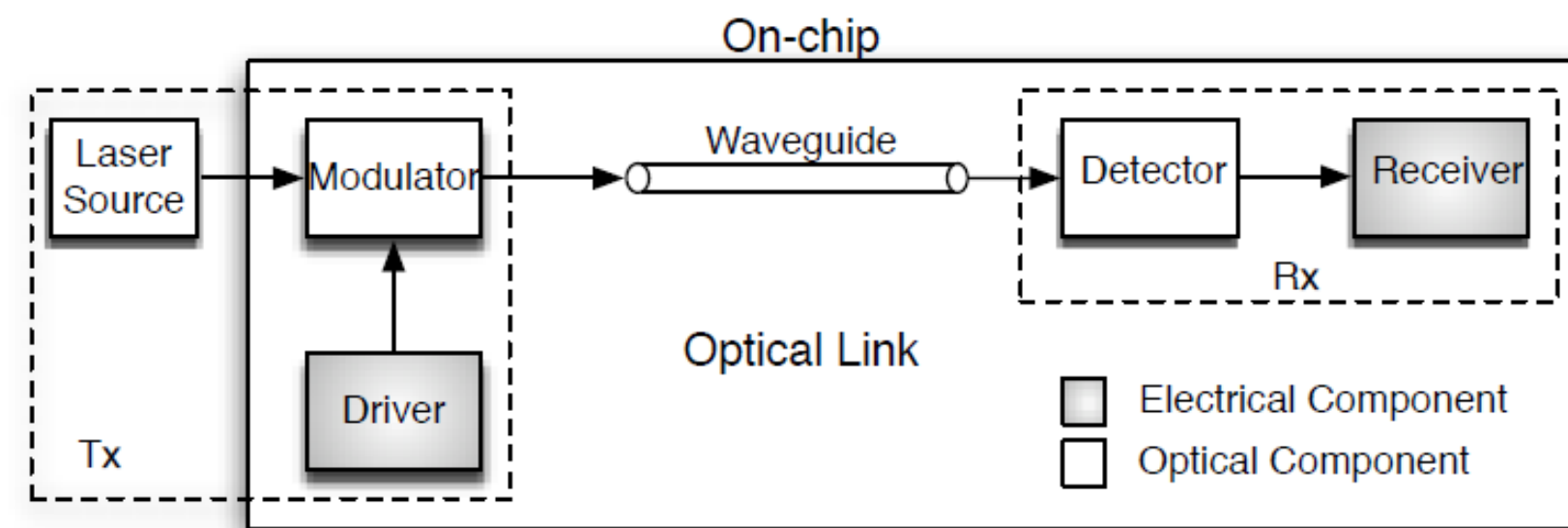
V : Voltage

MOTIVATION (2)

- Optical Modulators to convert data from electrical to optical signals



[WDM: image source](#)



[image source](#)

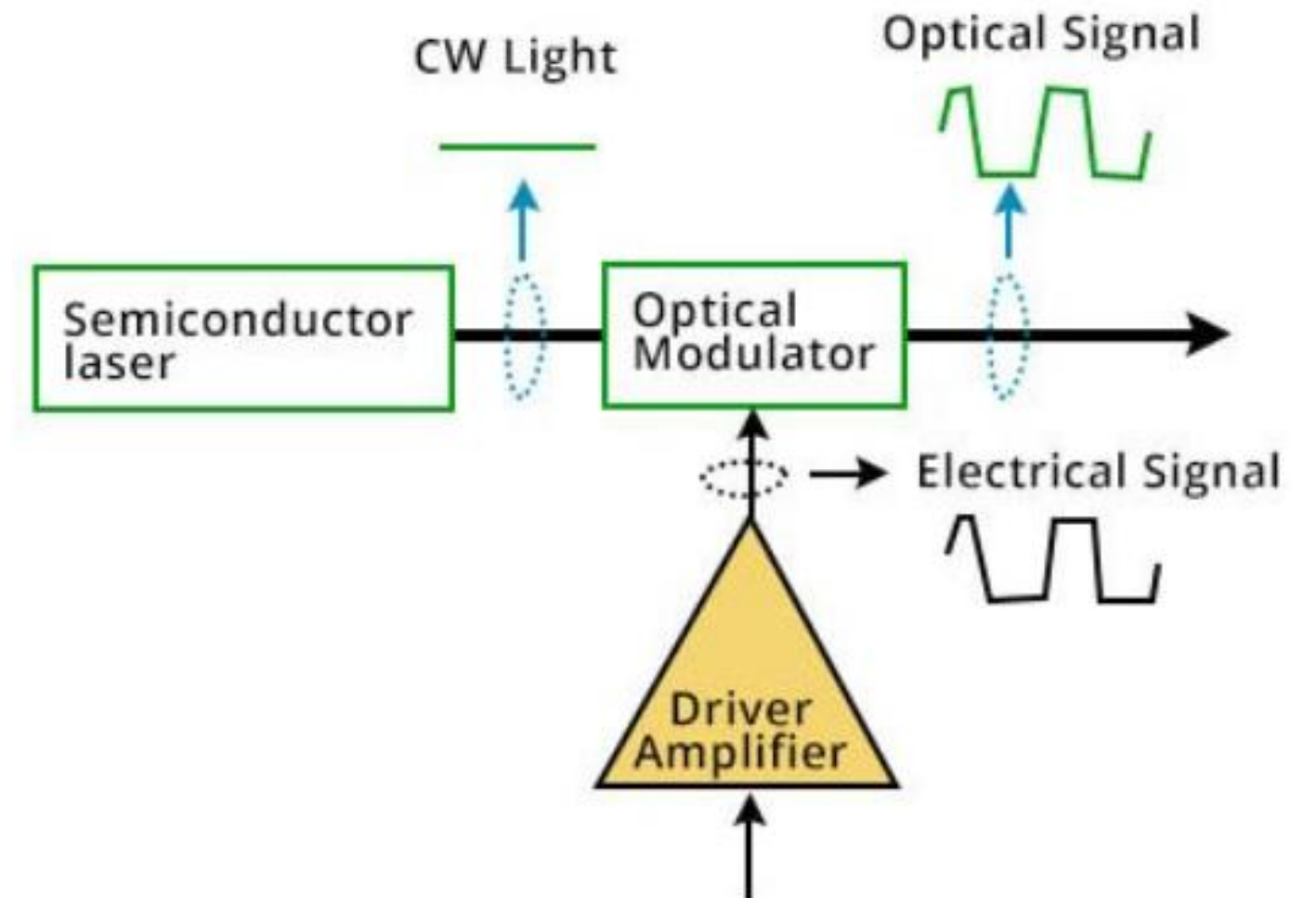
On-chip optical transmission block diagram

REQUIREMENTS

- Optical Modulators
 - Small
 - Low power
 - High efficiency
 - Low loss
 - High Speed
 - High Extinction ratio
 - Compatible with CMOS

OPTICAL MODULATORS PROFILE

- High speed switch for CW laser
- Electric field is applied → Change in optical properties
- The change in refractive index → characteristic of light passing through the device



MODULATOR TYPES

- MZMs are larger in size and consume high power ($>1\text{pJ/bit}$)
- Slow Light MZMs are short but has larger optical loss or narrow optical bandwidth
- Ring Modulators are compact and efficient
- Electro-absorption modulator (EAM)

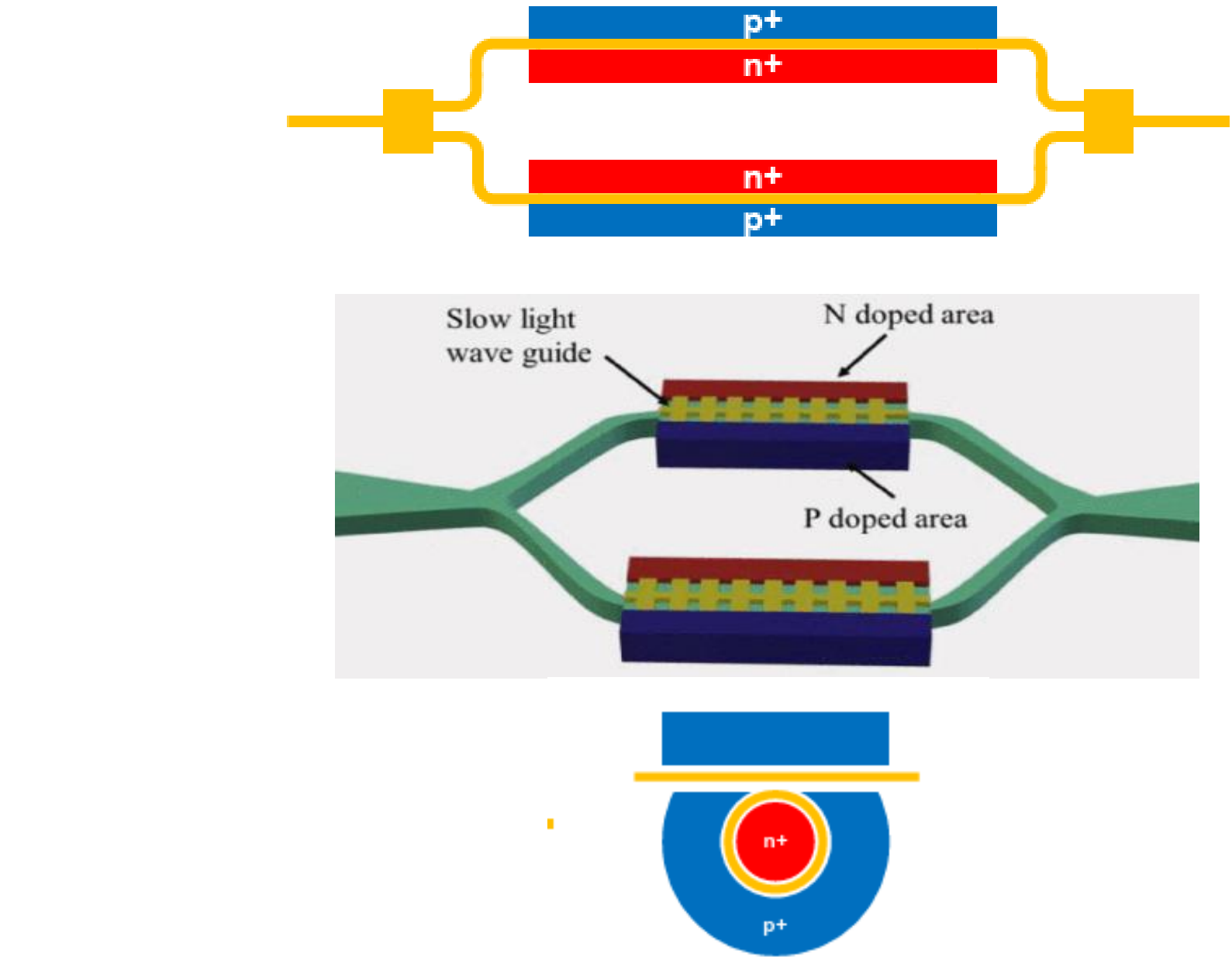
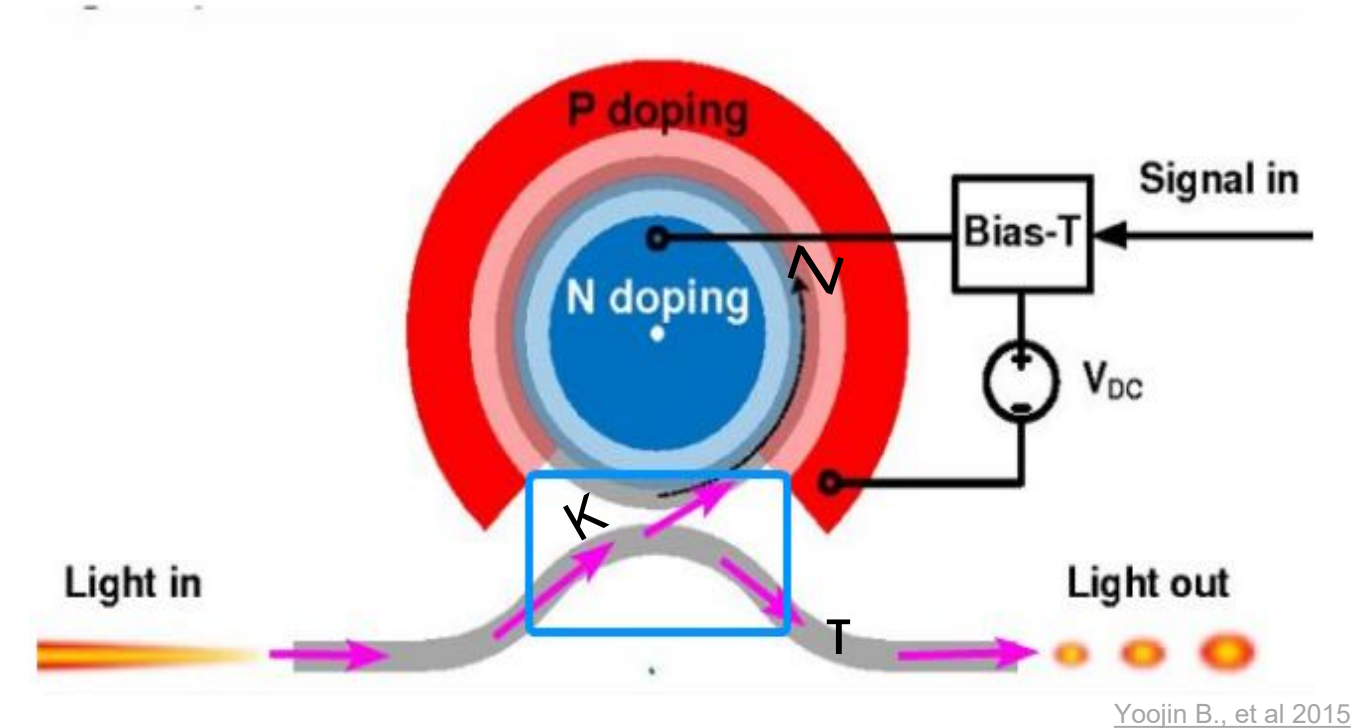


Figure of Merit	Mach Zehnder	Microring
Footprint	Large -	Small +
Extinction Ratio (ER)	Small -	Large +
Insertion Loss	High -	Low +
Wavelength Sensitivity	Low +	High -

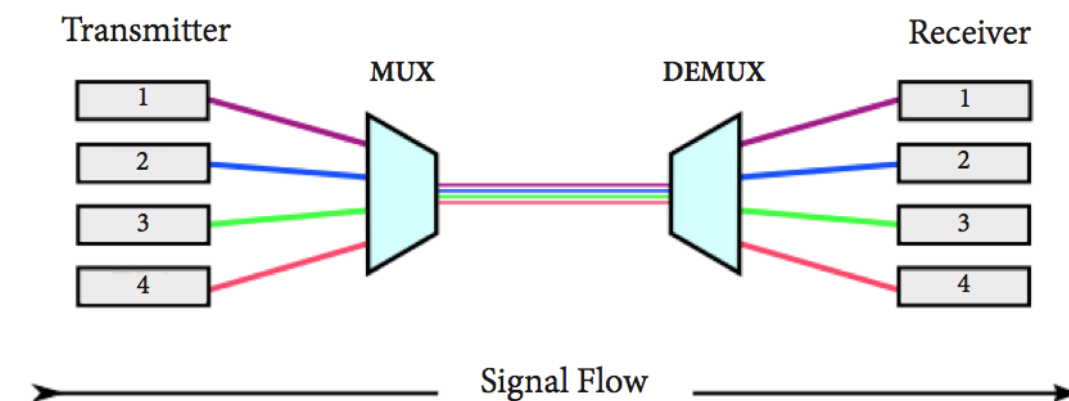
MOTIVATION (3)

Why Active Ring Modulators?

- Allows Wavelength Division Multiplexing (WDM)
- Precise Frequency selection
- Compact size and Efficient



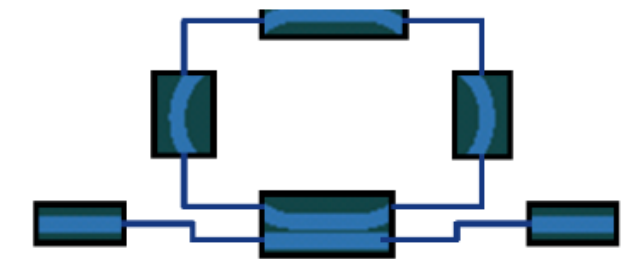
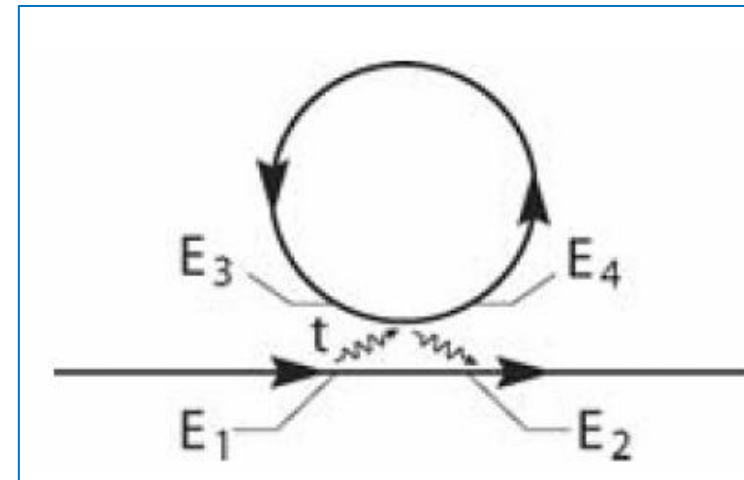
Wavelength Division Multiplexing (WDM)



precisionot.com

TRANSFER MATRIX METHOD

- Compound structures like ring modulators can be decomposed into individual constituent parts.
- Each part can be expressed in the form of a transfer matrix, giving the complex (amplitude and phase) relationships of "left-side quantities" to the "right-side quantities".
- This allows us to express a ring modulator as a multiplication of Transfer Matrix of each individual part(Interconnect).



Ansys

$$\begin{pmatrix} E_4(w) \\ E_2(w) \end{pmatrix} = \begin{pmatrix} r & jt \\ jt & r \end{pmatrix} \begin{pmatrix} E_3(w) \\ E_1(w) \end{pmatrix}$$

$$MZI_s = Y_r \Phi Y_f = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} e^{i\phi} & 0 \\ 0 & e^{-i\phi} \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}} \\ 1 \\ \frac{1}{\sqrt{2}} \end{pmatrix}$$

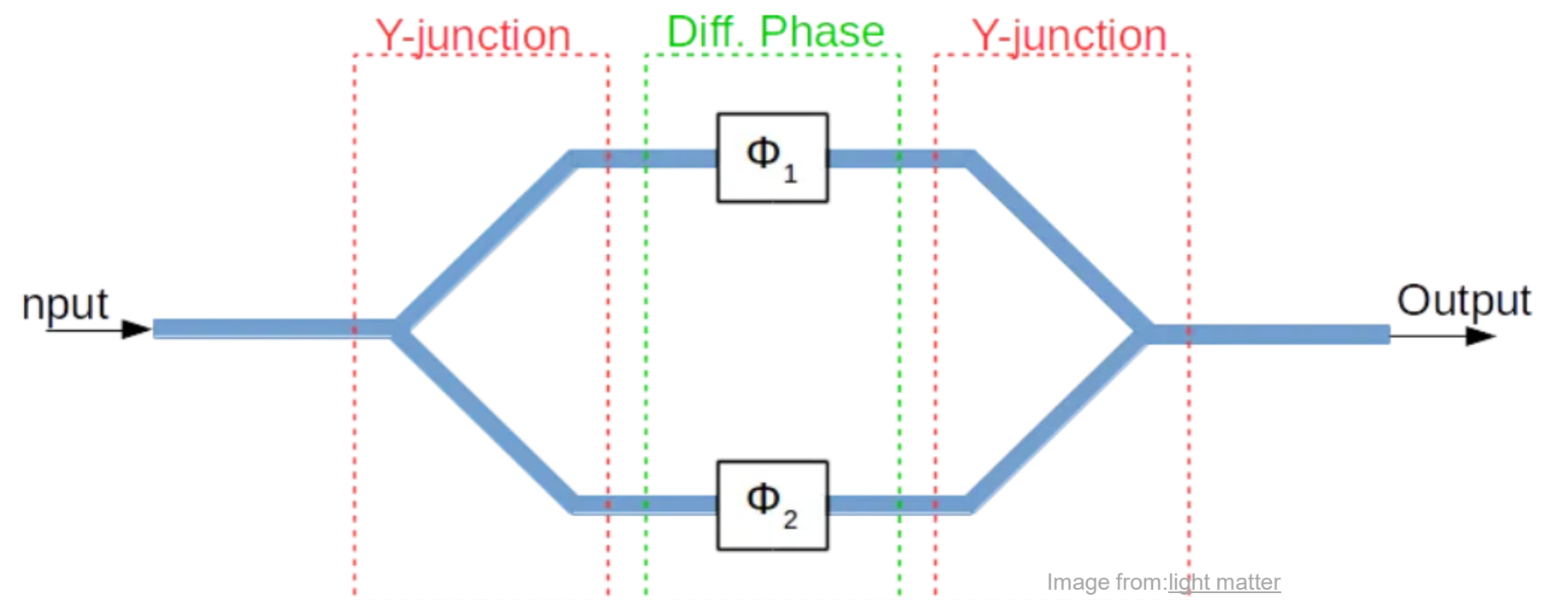


Image from: [light matter](#)

OPTICAL INTERCONNECTS IN DATACENTERS

- NVIDIA, Broadcom, and TSMC are now deploying these technologies.

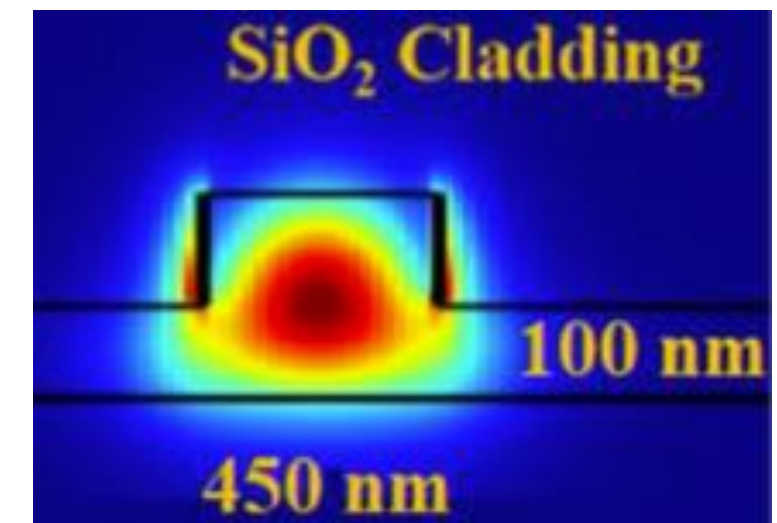
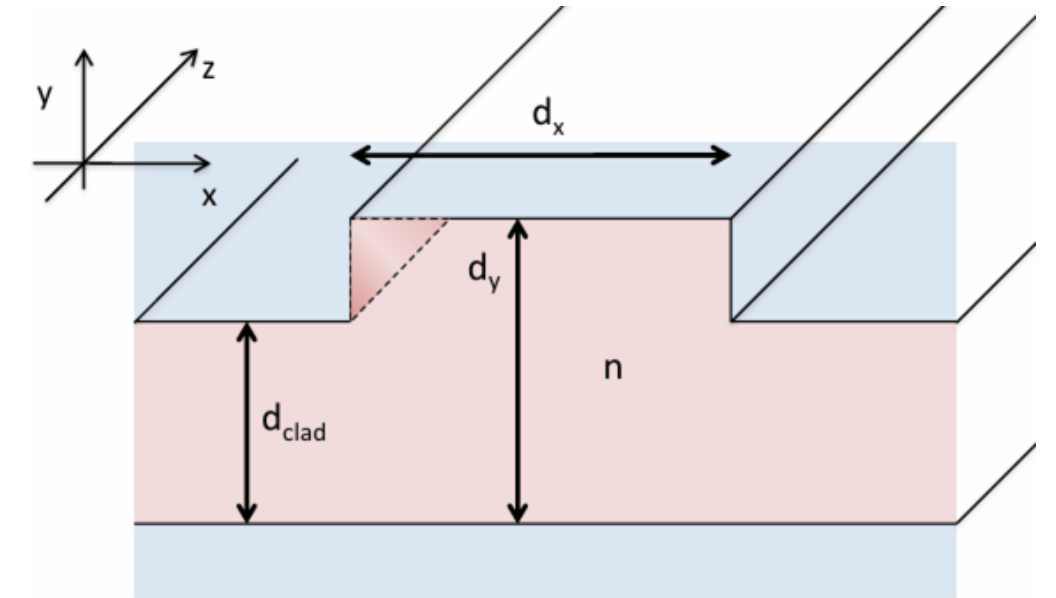


<https://spectrum.ieee.org/co-packaged-optics>

WAVEGUIDES

RIB WAVEGUIDE

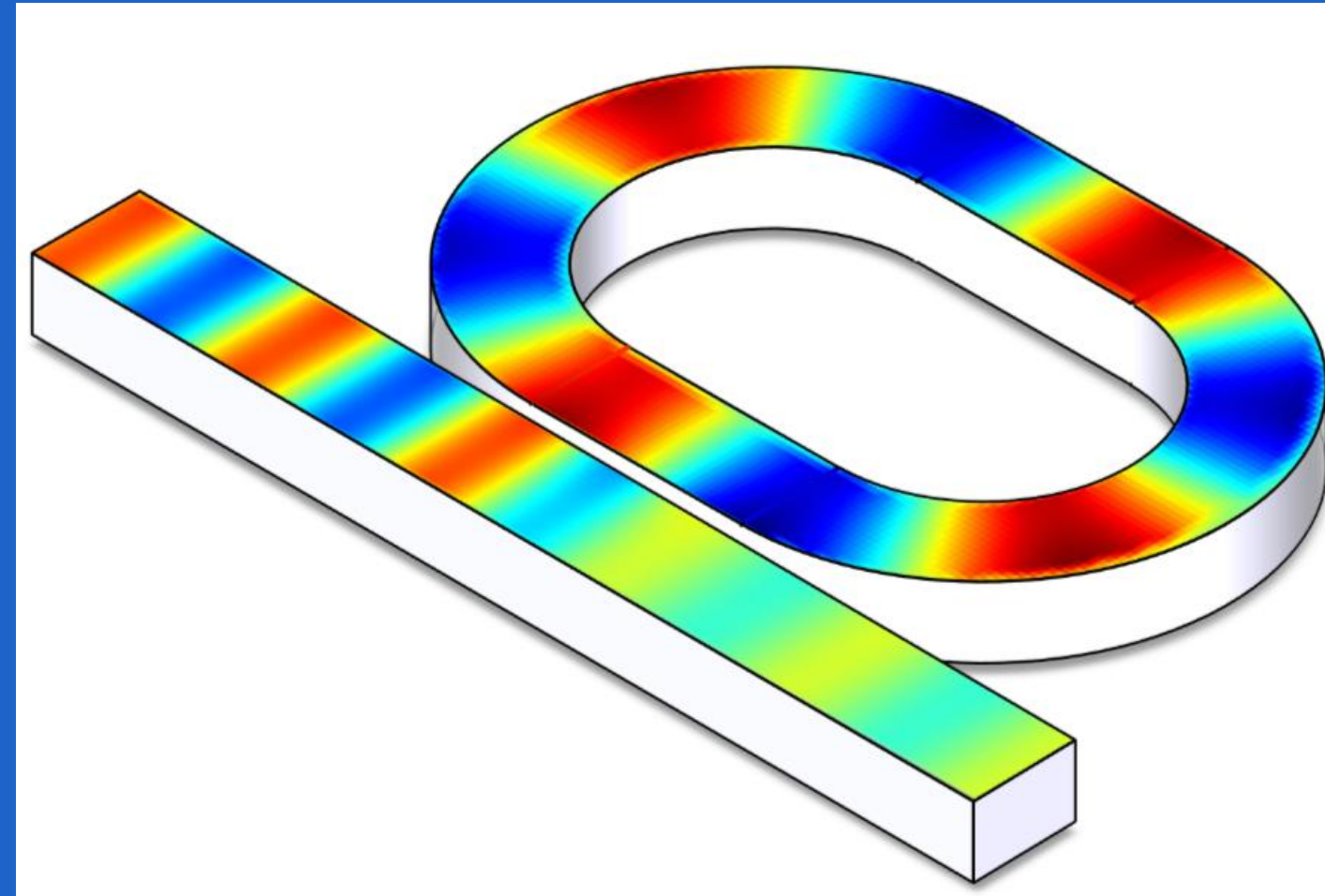
- Partially etched
- Less scattering loss than a strip waveguide
- A larger bending radius is required



BEND WAVEGUIDE

- Light is no longer fully confined
 - No TIR for some rays
 - wavefront revolves around center
 - Radiation losses increase with smaller bend radii
 - Strong field on the outer interface → scattering losses
 - As the bend radius decreases, losses and radiation increases

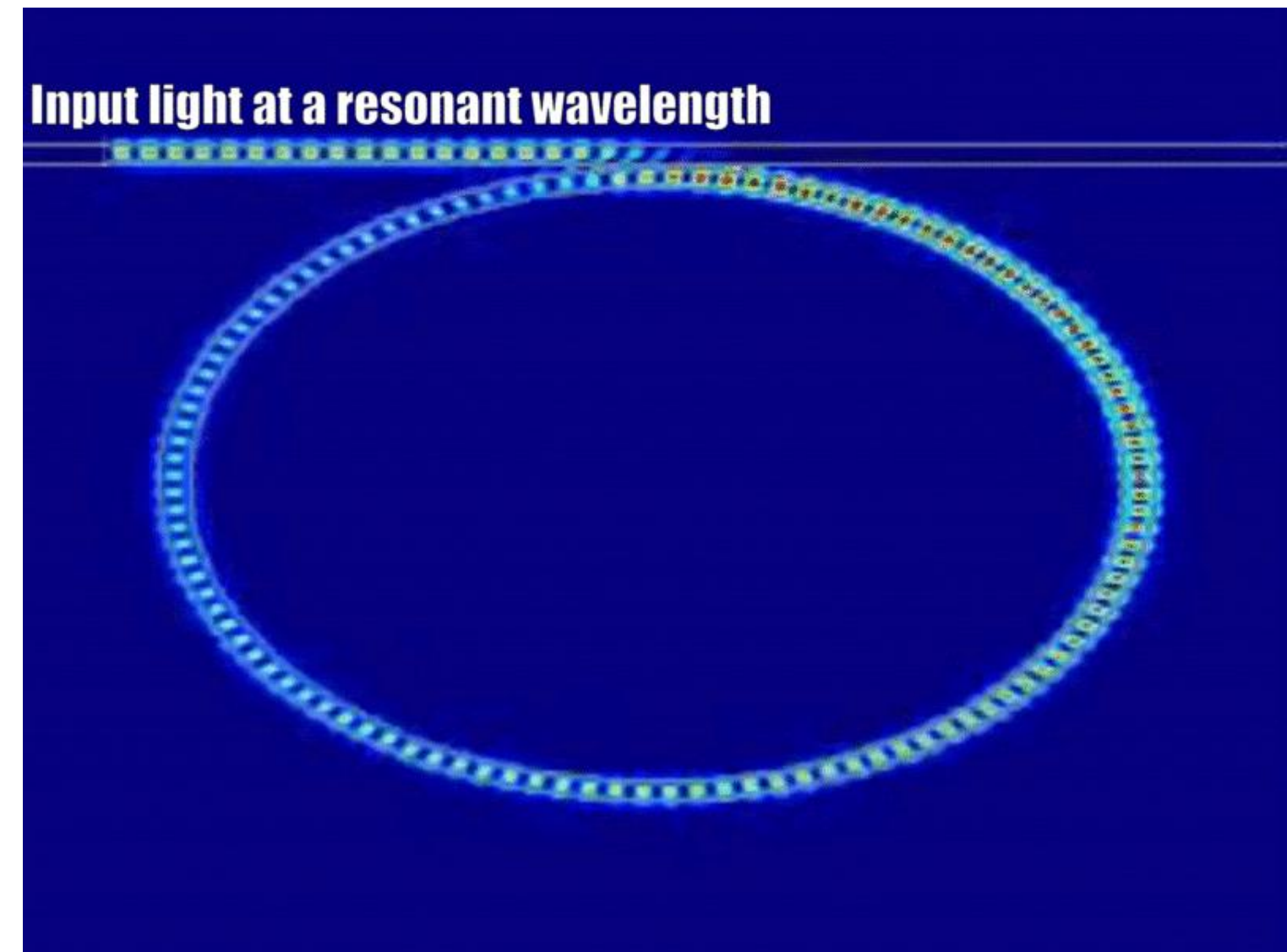




RING RESONATORS

RING RESONATORS

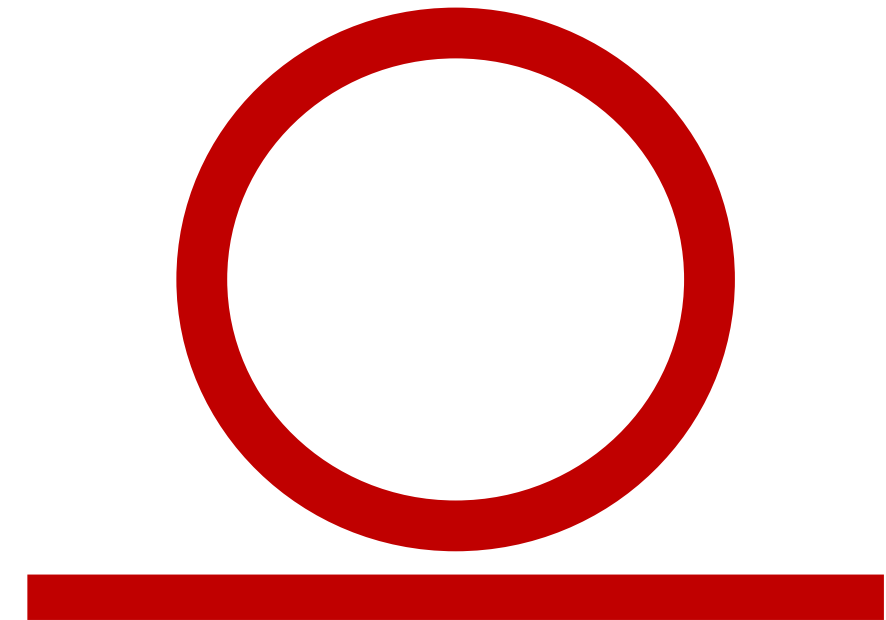
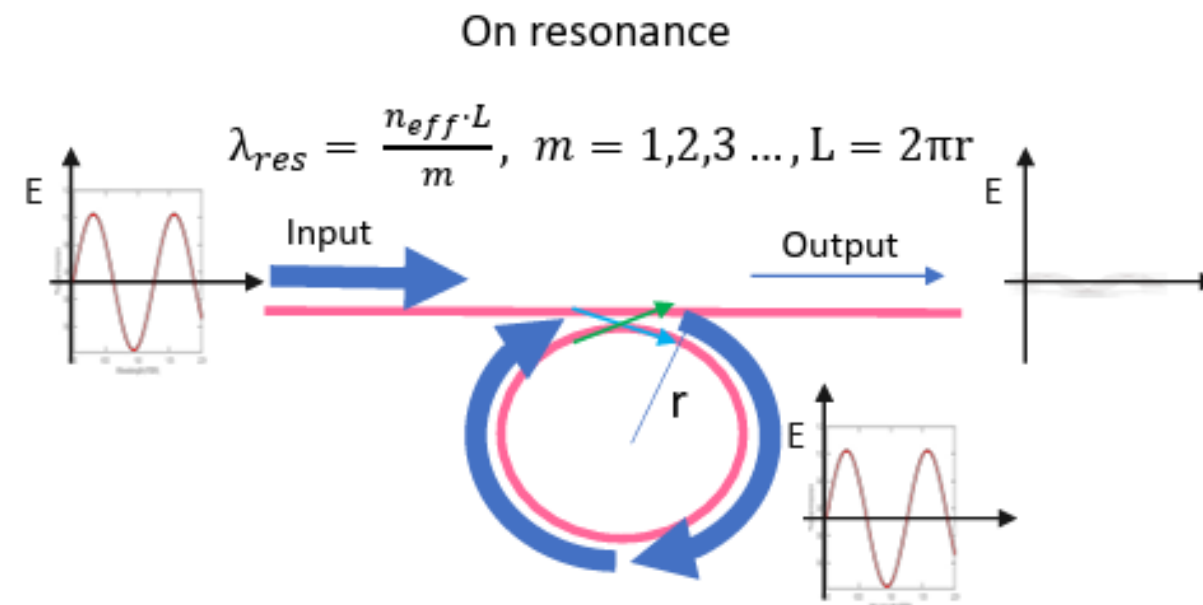
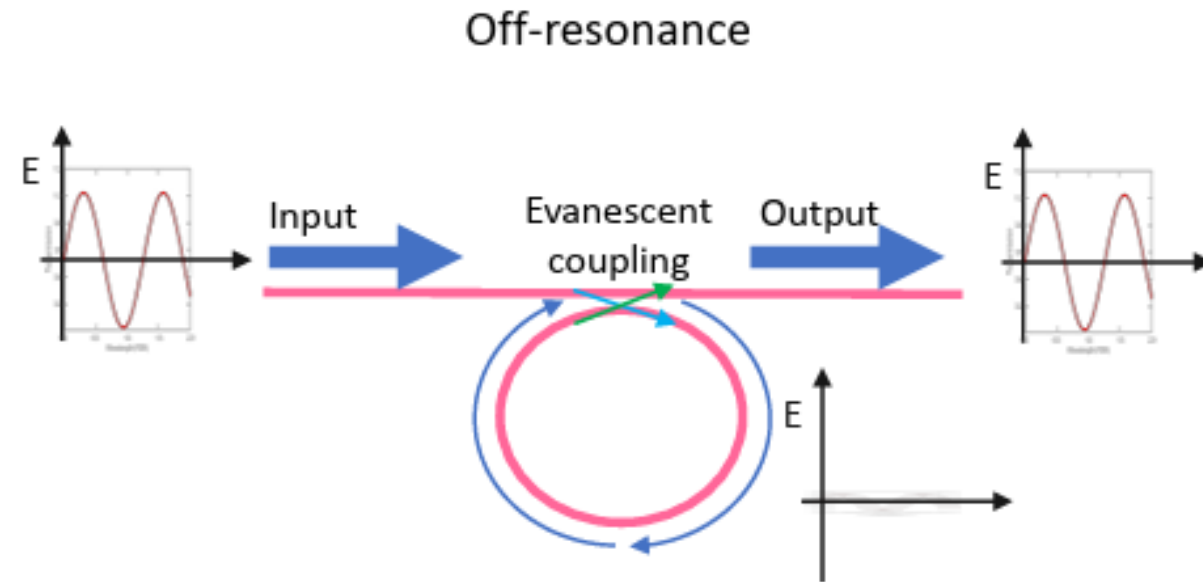
- Waveguide as a loop next to waveguide
- Roundtrip phase shift equals an integer times 2π
- Wavelength selective device



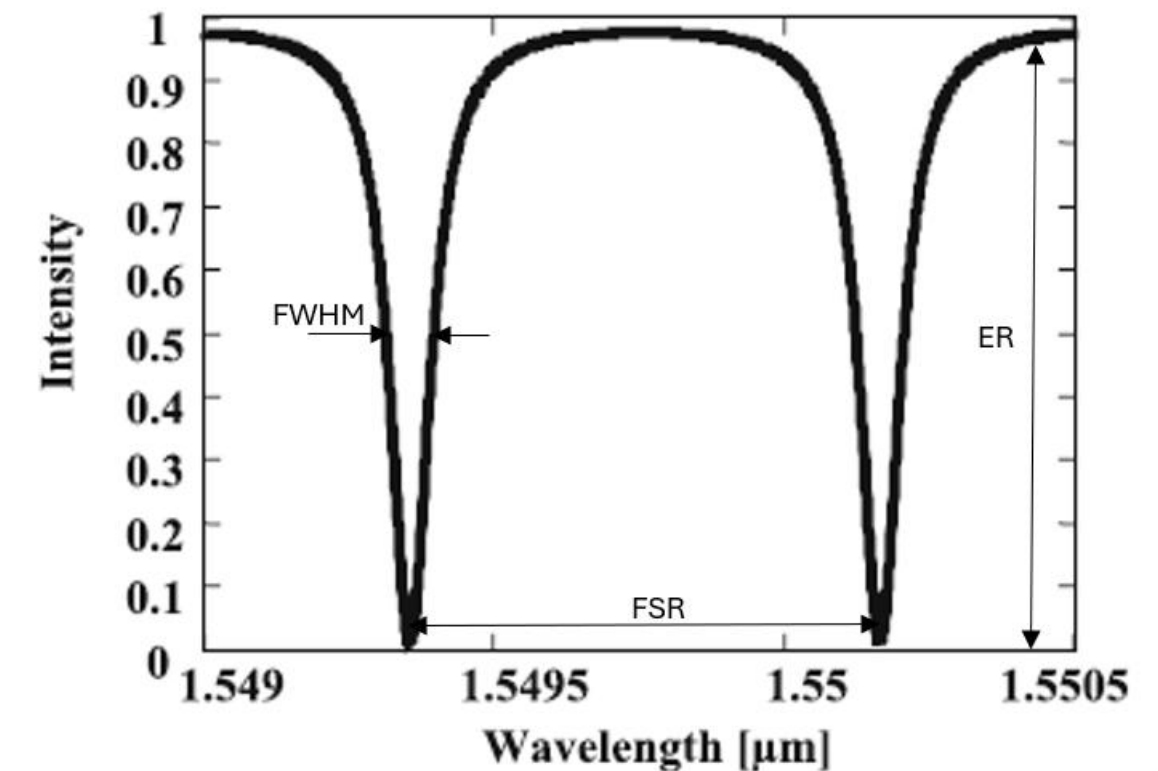
$$\lambda_0 = \frac{n_{eff}L}{m}, m = 1, 2, 3, \dots$$

ALL-PASS FILTER

- Dips around the ring resonances

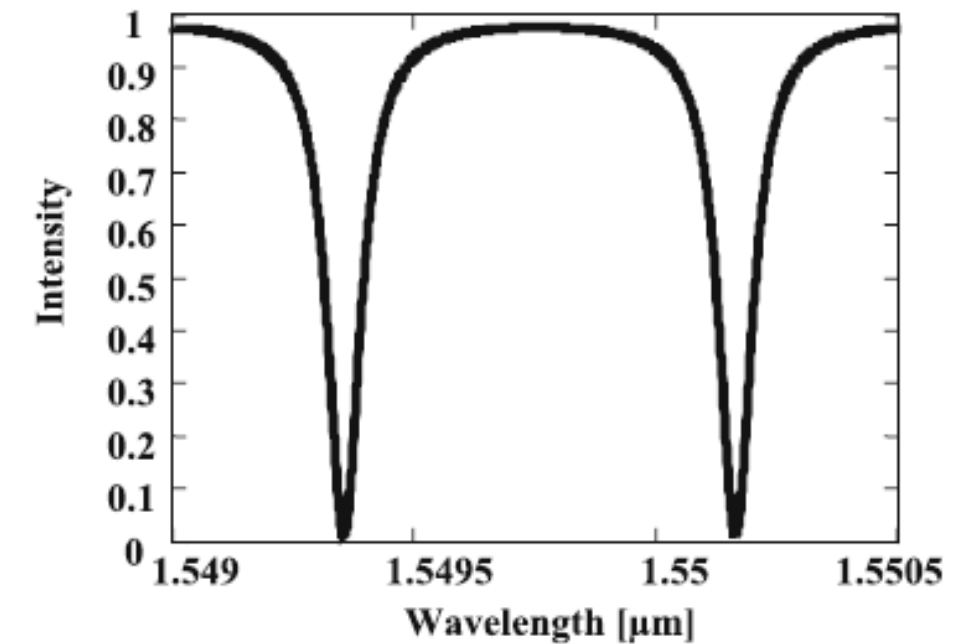


All-pass



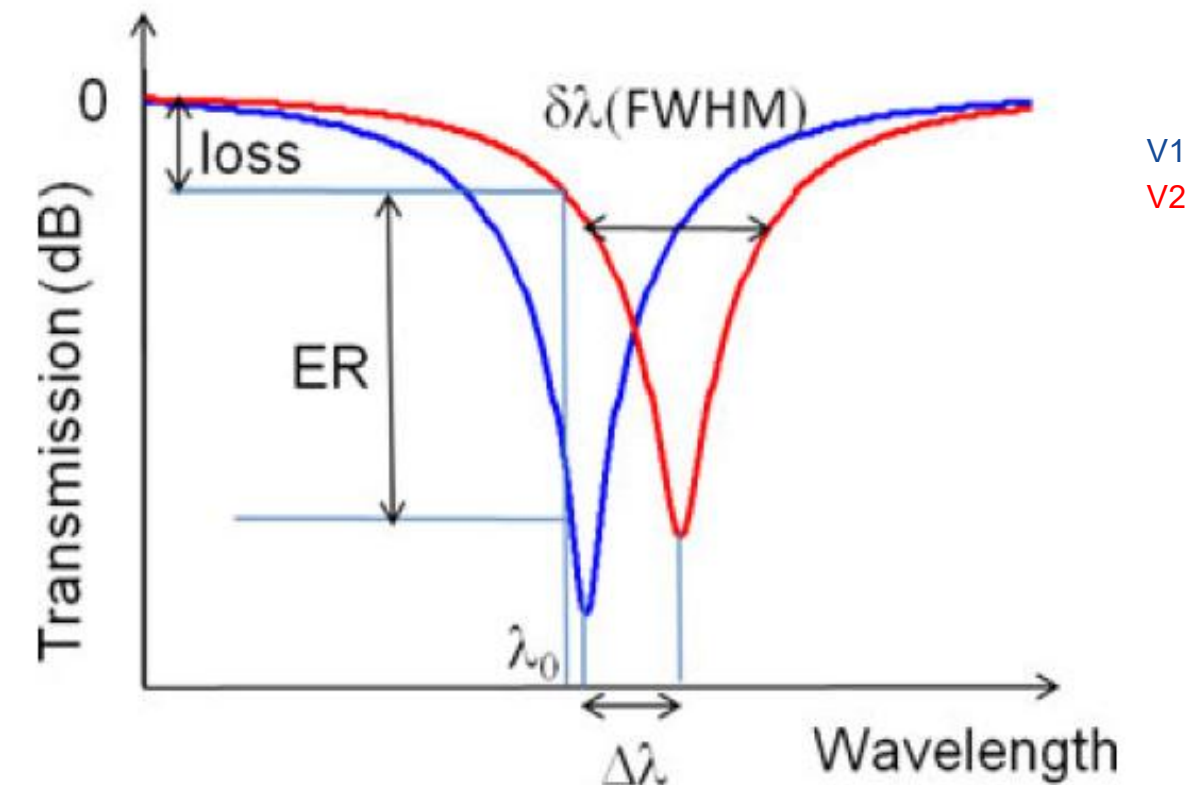
RING DESIGN

- Single Mode waveguide
- Minimize waveguide bending loss
- Maximize mode overlap with core region



$$Q = \frac{\lambda}{\delta\lambda} \approx \frac{n_g \pi}{\lambda \alpha}$$

To maximize ER maximize $\frac{\Delta\lambda}{\delta\lambda}$



QUALITY FACTOR

- Quality Factor: Measure of sharpness of the resonance relative to its central frequency

$$Q = \frac{\lambda_{res}}{FWHM}$$

$$Q = \omega_0 \frac{\textit{field energy stored by resonator}}{\textit{power dissipated by resonator}}$$

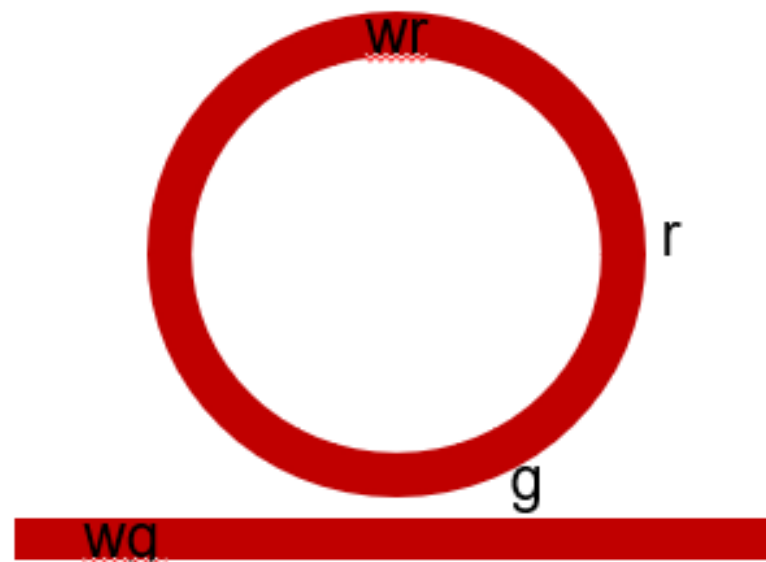
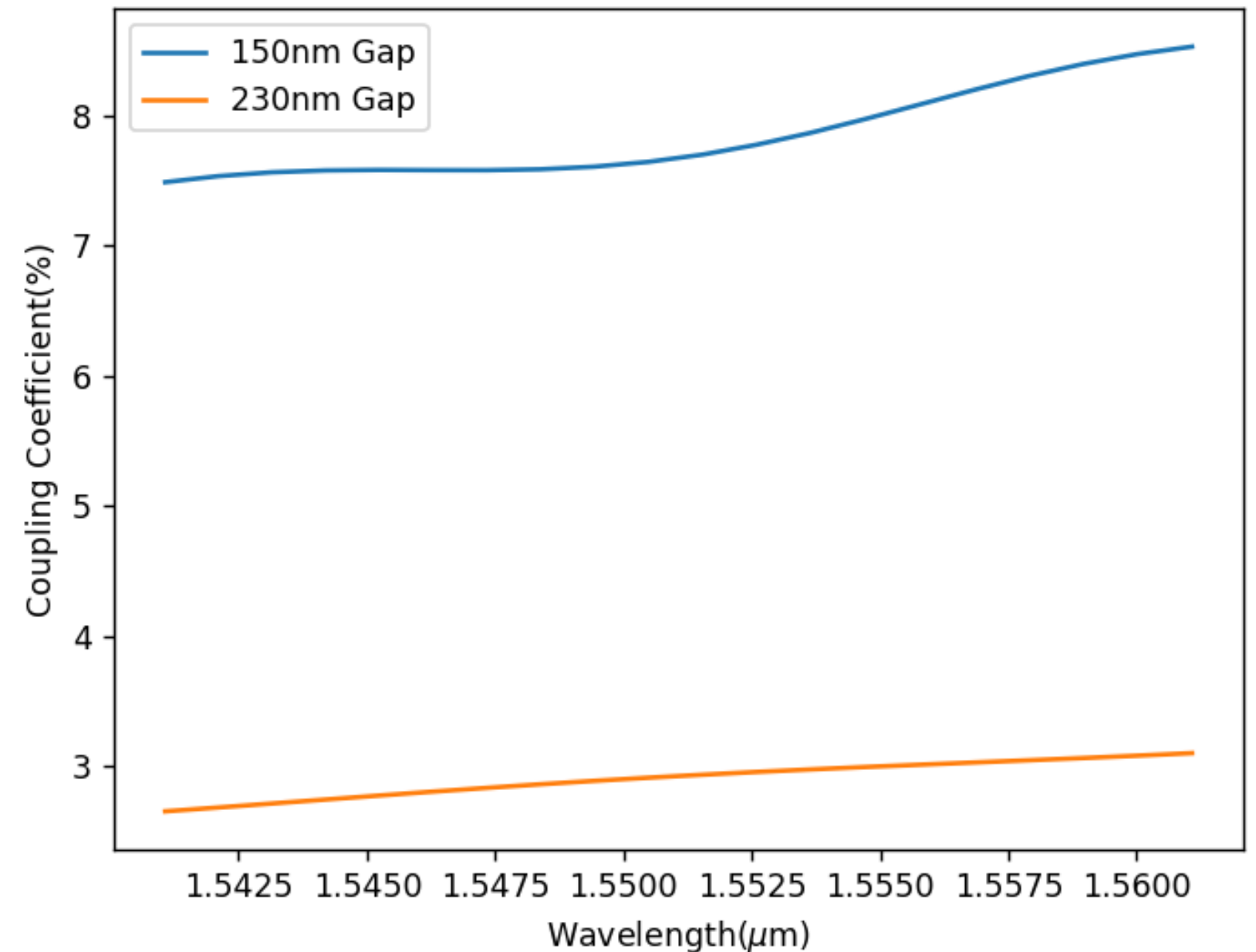
Q-FACTOR VS GAP

$$\frac{1}{Q_{loaded}} = \frac{1}{Q_{intrinsic}} + \frac{1}{Q_{external}}$$

Trade-off for coupling gap:

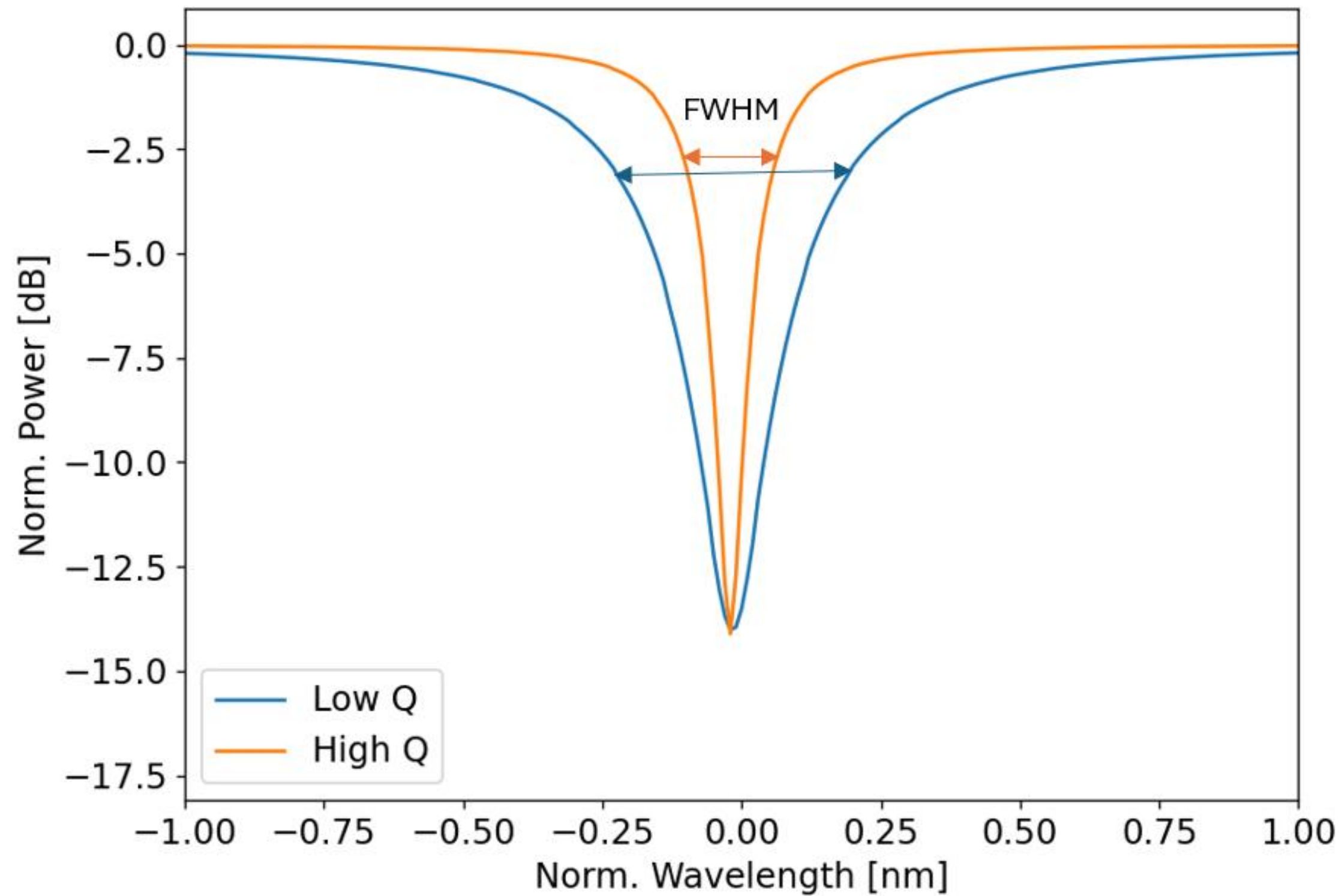
Small gap → More coupling → More light escaped → Low Q factor

Large gap → Less coupling → Less light escaped → High Q factor



$$Q \approx \frac{\pi n_g L}{\lambda} \frac{1}{K^2}$$

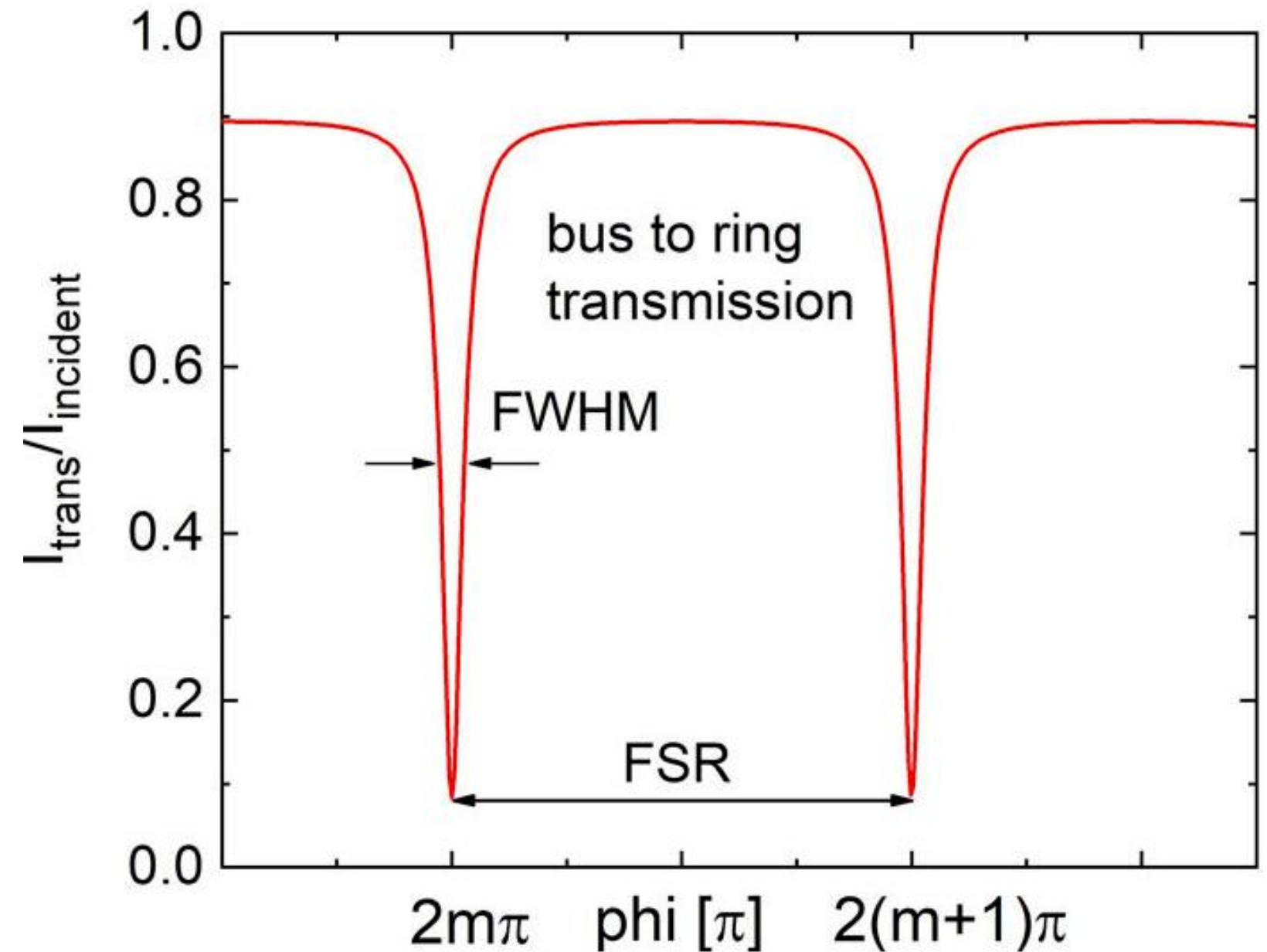
Q-FACTOR VS BANDWIDTH



FSR

- Multiple resonances
- Spacing between resonances
- Large FSR $\rightarrow$ small length

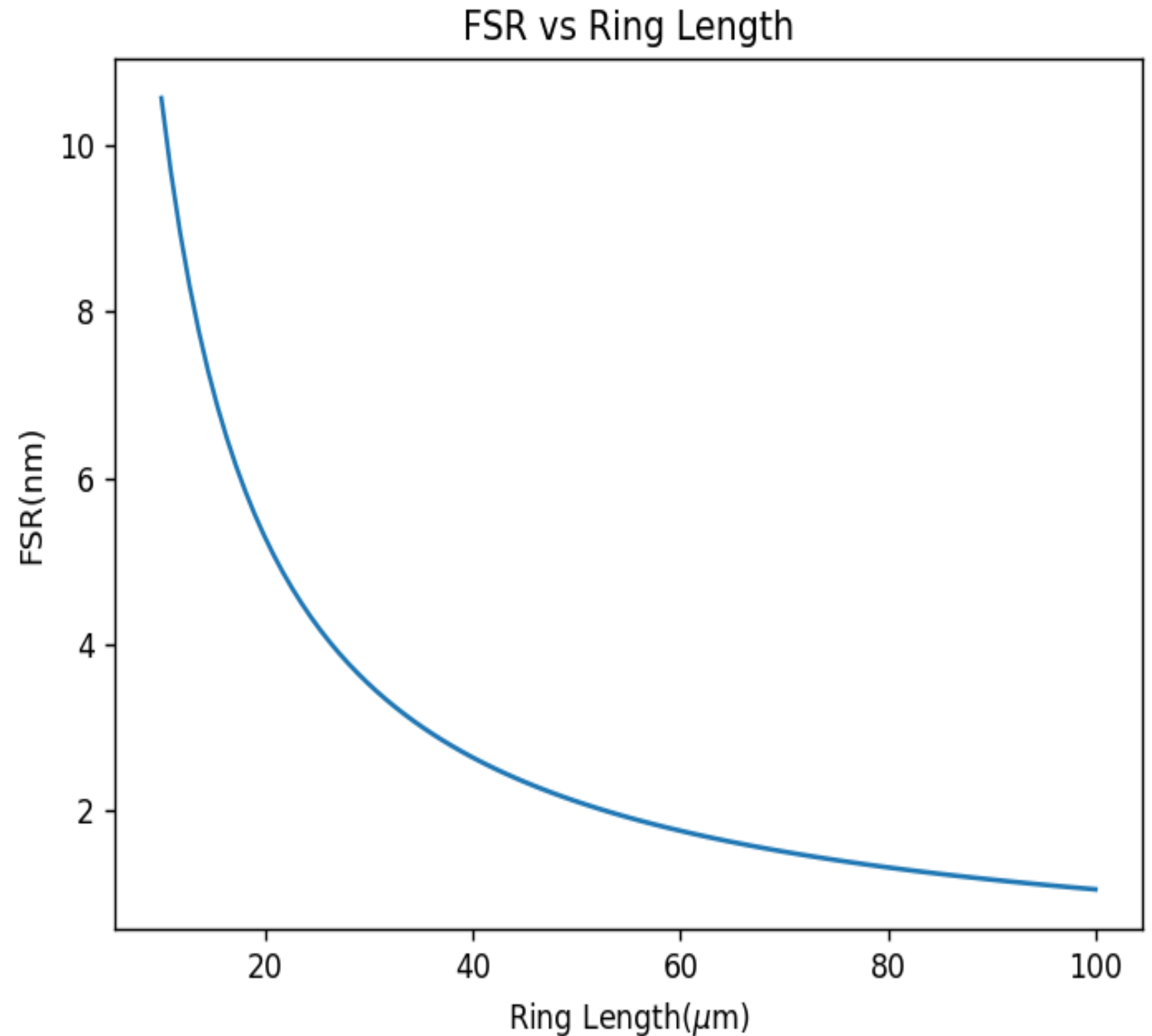
$$FSR = \frac{\lambda^2}{n_g L}$$



FSR

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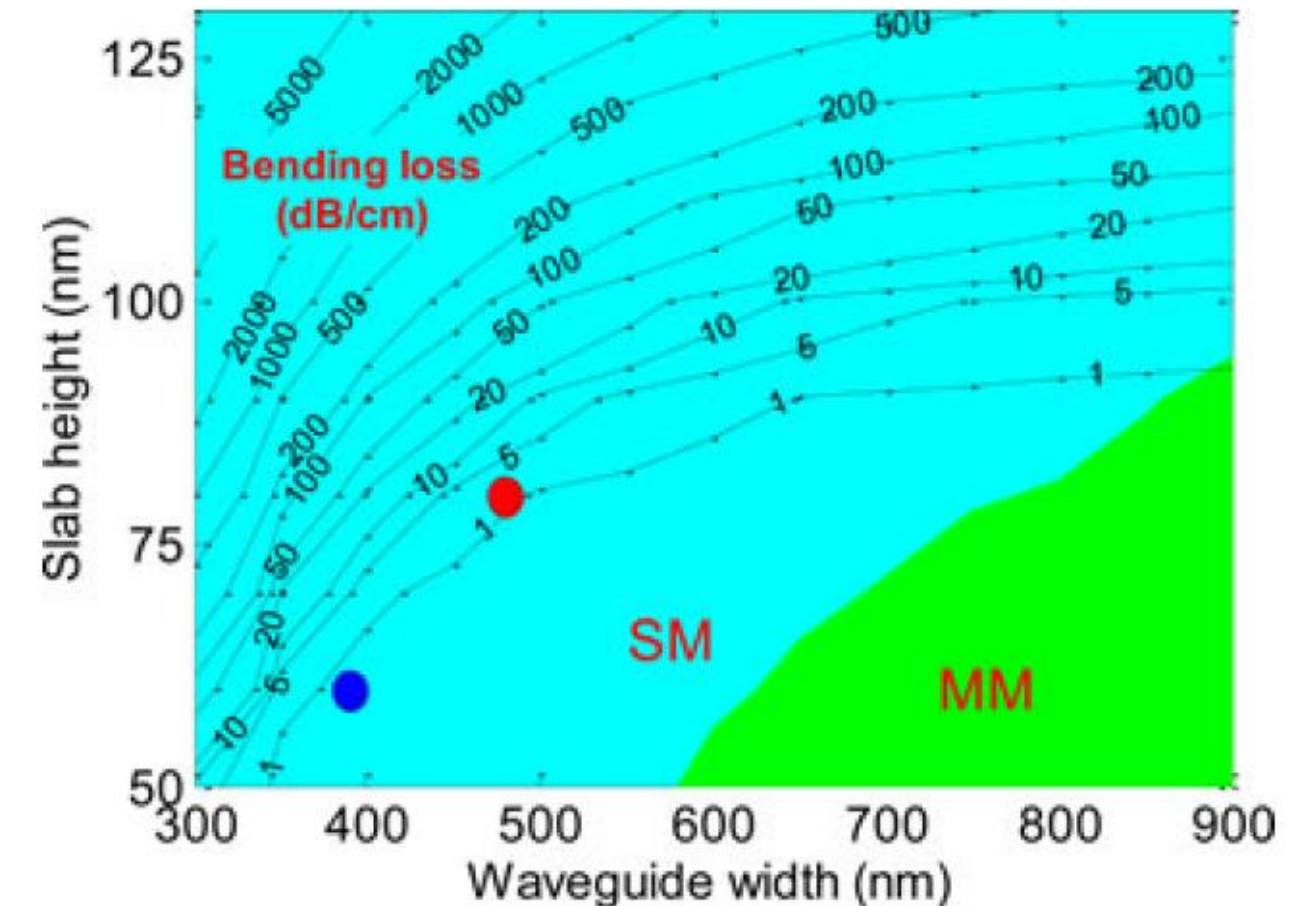
FSR VS BENDLOSS

- Smaller ring has large FSR
- Needs deeper waveguide etching for low bending loss
- Deeper etch leads to high resistance
- RC-limited bandwidth
- Compromise on radius and slab thickness

$$\frac{1}{f_{3dB}^2} = \frac{1}{f_Q^2} + \frac{1}{f_{RC}^2}$$

FSR VS BENDLOSS

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- Needs deeper waveguide etching for low bending loss
- Deeper etch leads to high resistance
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EXTINCTION RATIO

- Ratio of optical power in ON and OFF state
- Coupling coefficient affects the ER
- Critical coupling gives high ER as the destructive interference occurs at the output
- Under coupling or over coupling reduces ER

OVERVIEW

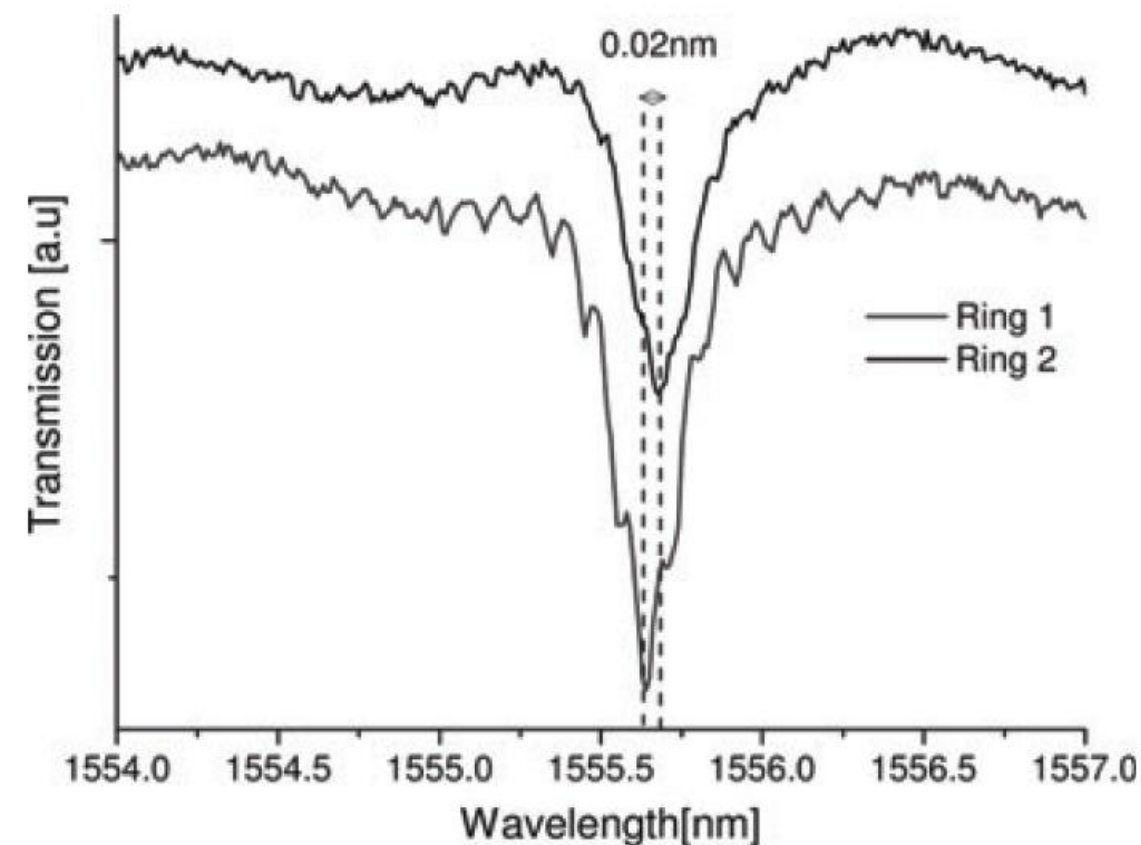
Metrics	Influenced by	Design Parameters
FSR	Length of the ring	Radius of the ring
Q Factor or FWHM	Losses in the ring	Radius, Gap, Slab thickness
Extinction Ratio	Coupling coefficient	Gap

RING RESONATORS

- Operating wavelength of fabricated rings differ from their targets

- ✓ Compact Footprint
- ✓ Low power consumption

- x Highly sensitive to process variations
- x Thermal instability

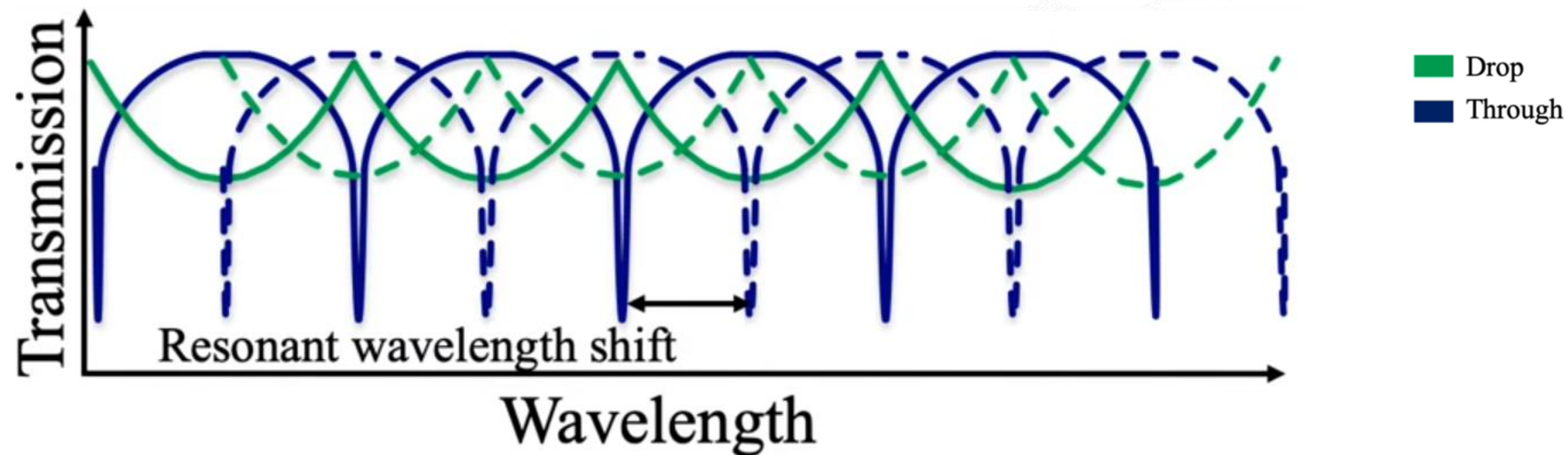
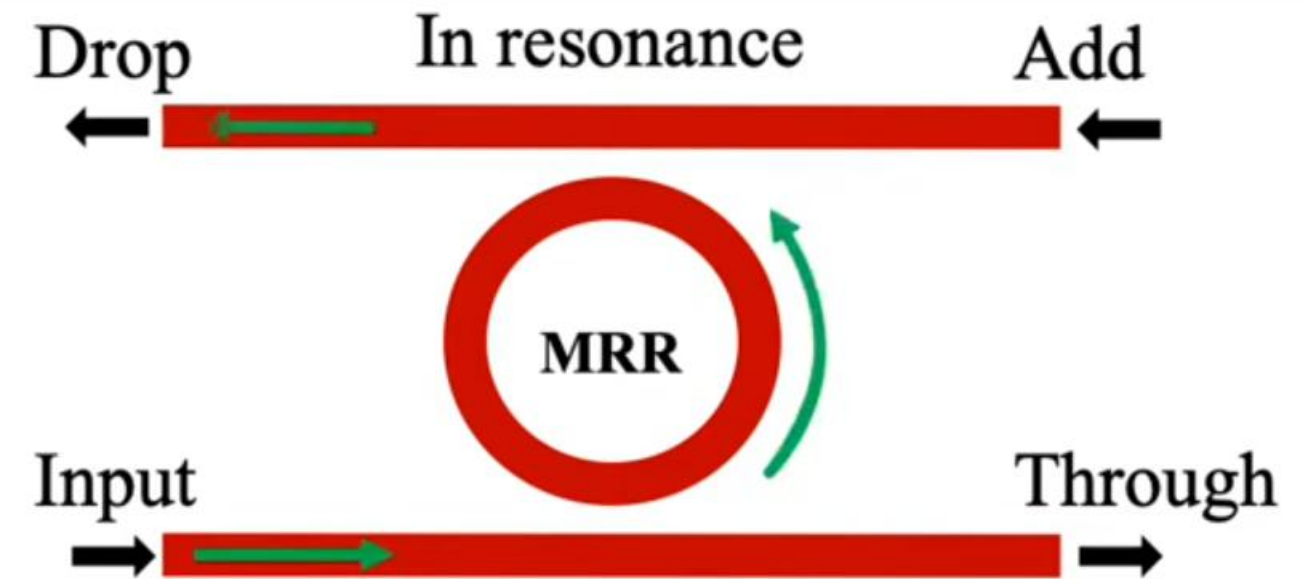


- ✓ Filters
- ✓ Modulation
- ✓ Switching

Spectral response of identical APF rings within a chip

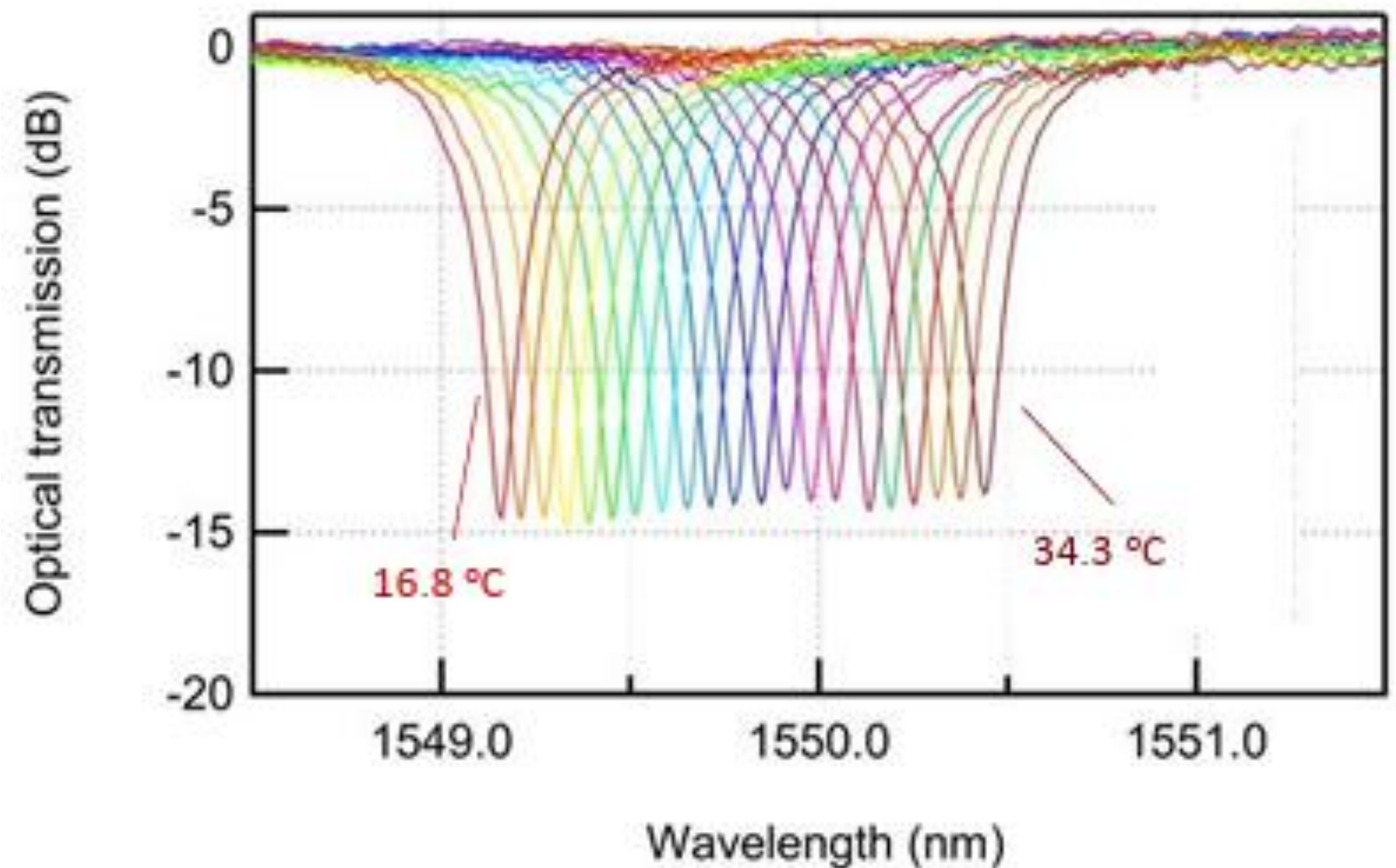
PROCESS VARIATIONS

- Due to small size and dependence on coupling
- High impact of thickness variability



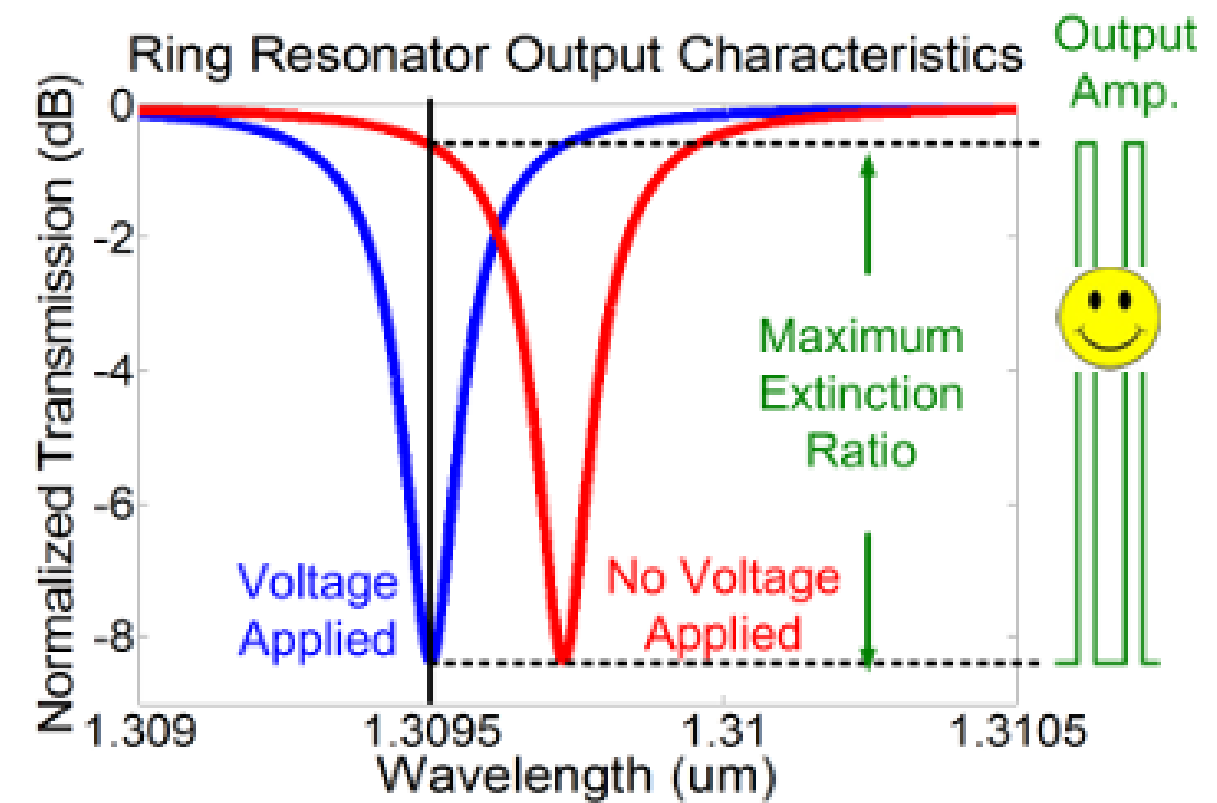
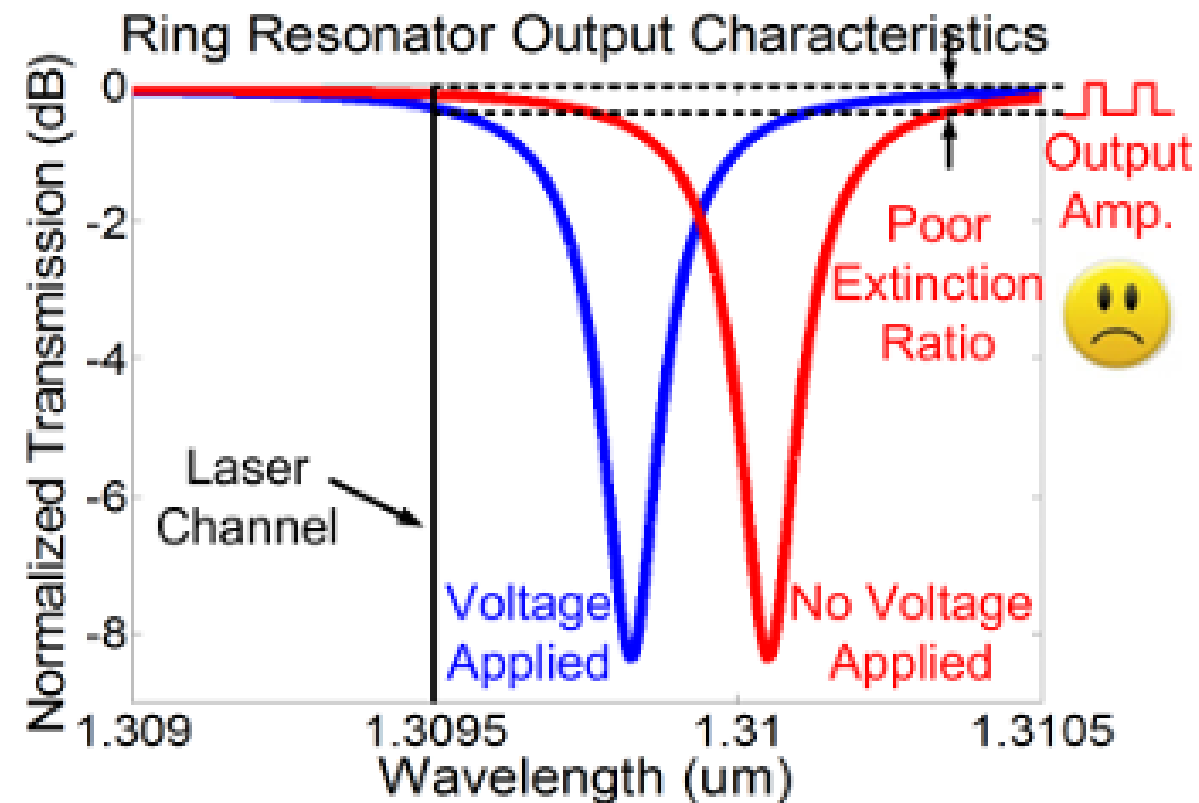
THERMAL INSTABILITY

- Silicon has a large thermo-optic coefficient
- Refractive index changes significantly with temperature
- Resonant wavelength changes with temperature
- Thermal stabilization is needed



ER IMPACT

- Small dimensional errors leads to low ER



After Tuning

THERMO-OPTIC EFFECT

- Heater on the ring
- Change in effective refractive index
- Slow modulation $\sim 1\text{kHz}$ to 100kHz
- Speed is limited by thermal time constant
- Etching the substrate increases the efficiency

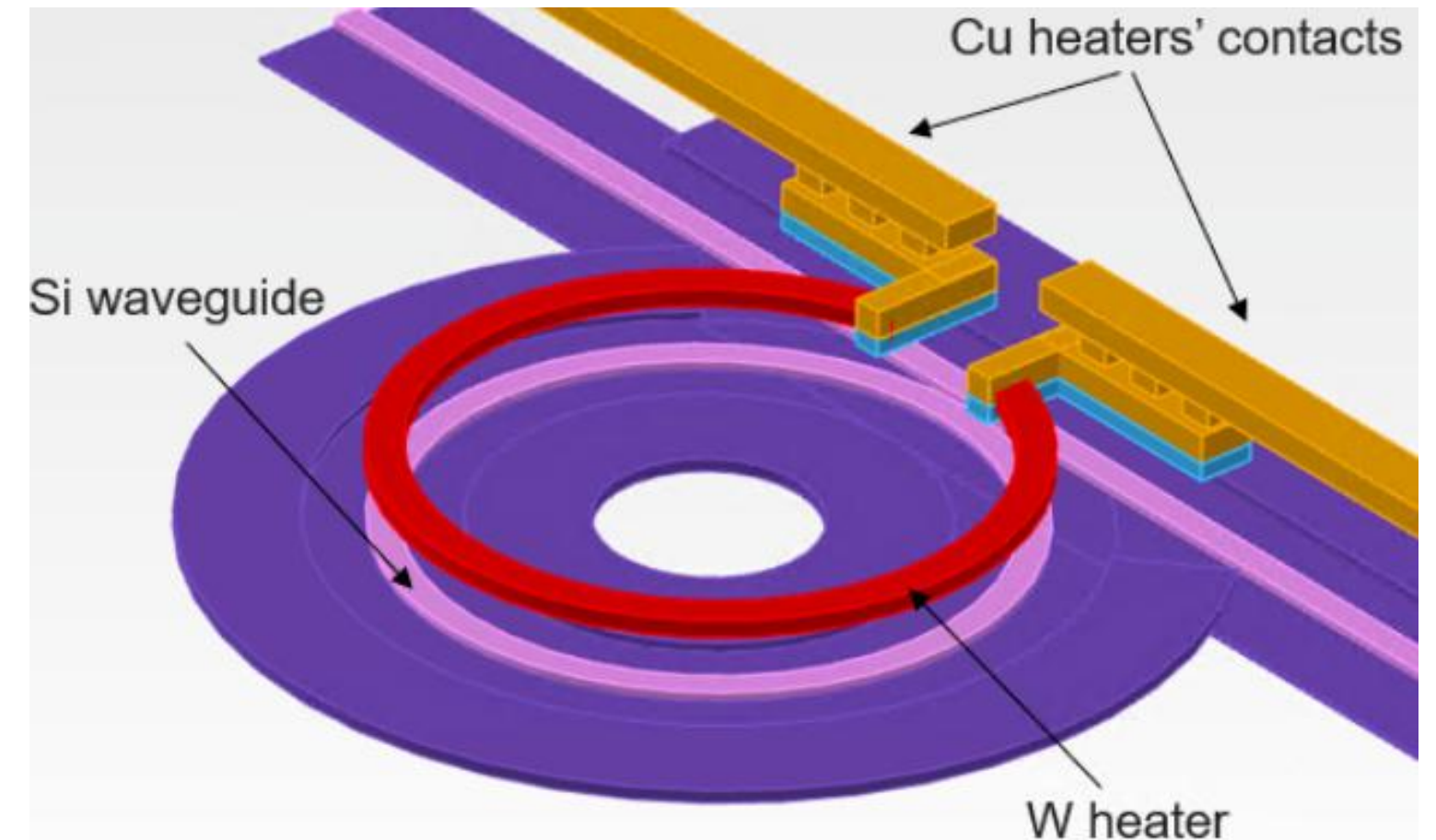


Image from: [Thermal annealing in silicon photonics ring modulators - IOPscience](#)

THERMO-OPTIC VS FREE CARRIER EFFECT

Feature	Heater	Carrier Injection
Physical Effect	Thermo-Optic Effect (dn/dT)	Plasma Dispersion Effect (Δn vs ΔN , ΔP)
Control	Heating	Voltage
Max. Speed	Slow (kHz to low MHz)	Fast (>10GHz range)
Limiting Factor	Thermal time constant	RC Time constant, Photon Lifetime
Primary Use	Tuning	Encoding high-bandwidth data signals

MODULATORS

MODULATION METHODS

- Thermo-optic effect:
Refractive index of material depends on temperature.
Slow ~kHz. Normally used for fine tuning of deviations in fab.
- Electro-optic (Pockels) effect:
Refractive index of material changes with external electric field.
Fast ~GHz, only works with non-centrosymmetric crystals.
- Electro-absorption (Franz-Keldysh) effect:
Only works for a very narrow wavelength band.
- Free carrier effects:
Refractive index of waveguide is modified by a change in free carrier concentration.
 - Injection
 - Depletion
 - Accumulation

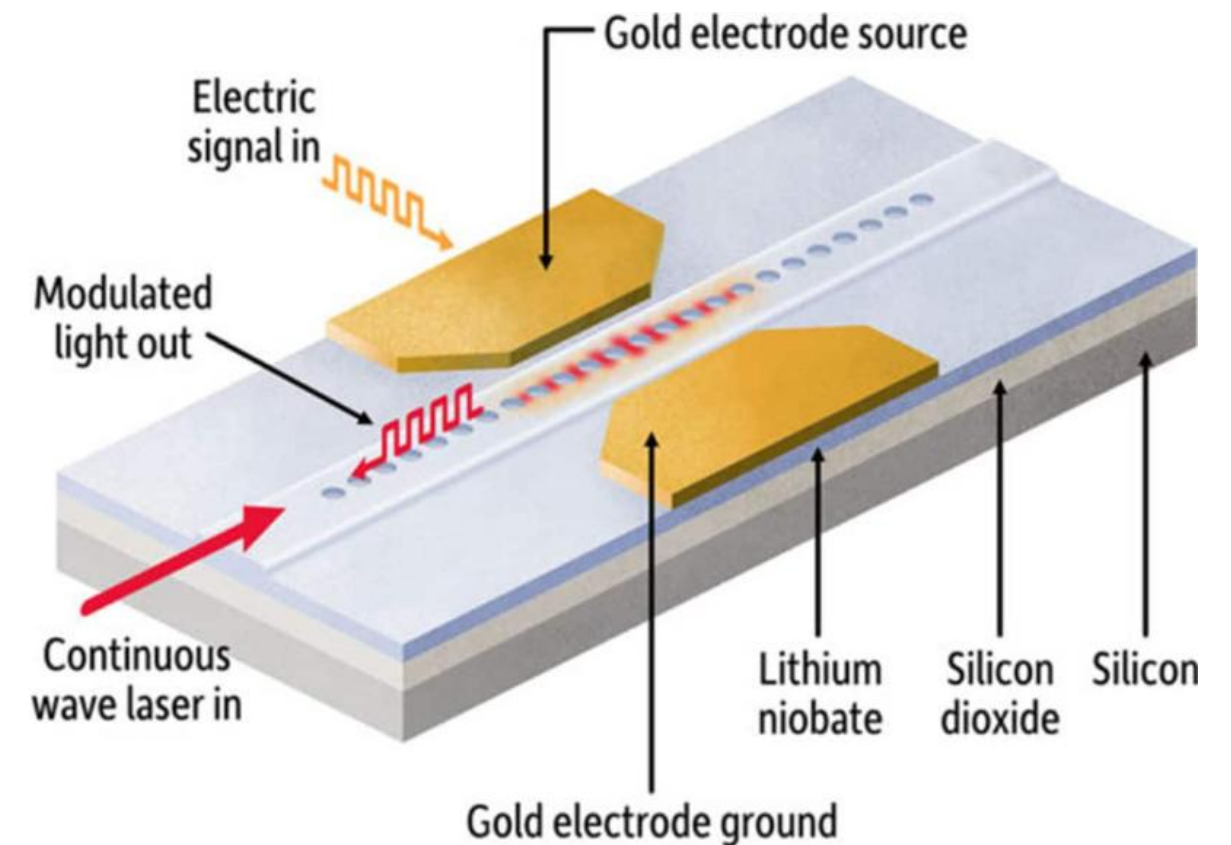
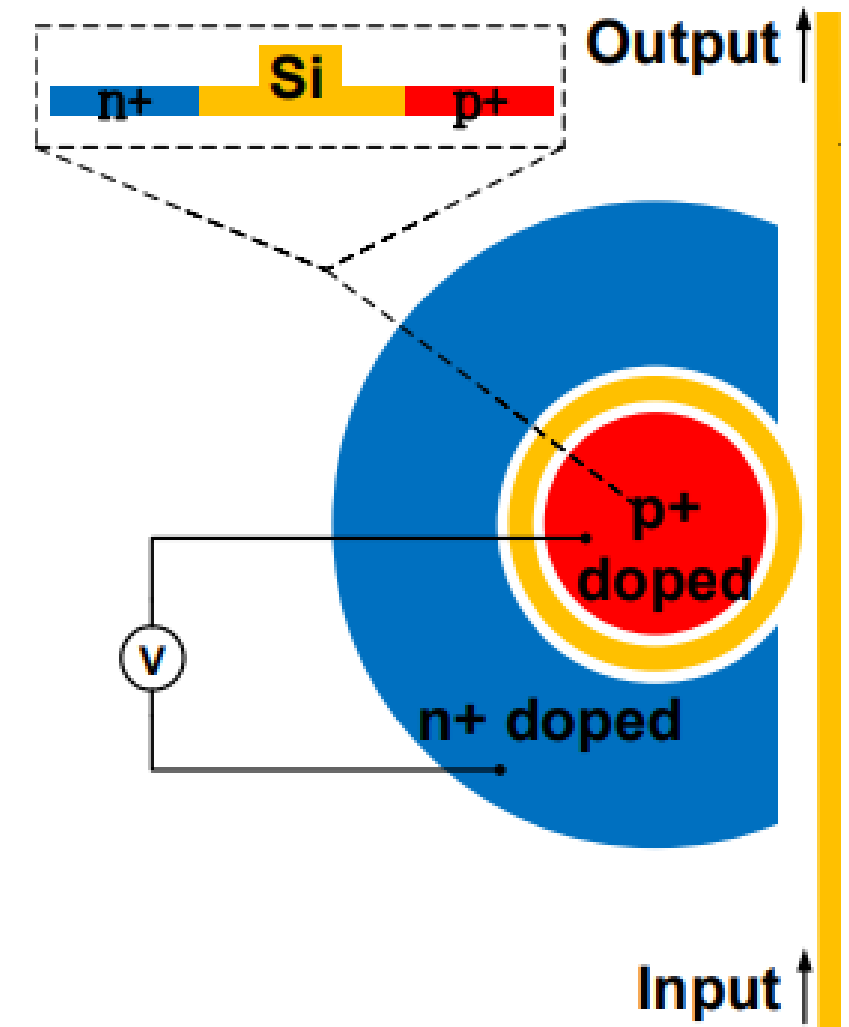


Image from: [Controlling light as it moves through integrated circuits](#)

FREE CARRIER CONCENTRATION EFFECT

- Resonator ring is doped to form diode-like structures
- Applying bias voltages to inject, withdraw or accumulate electrons & holes:
Density of carriers increases with the applied voltage.
- Plasma Dispersion effect:
Changing the carrier concentration changes refractive index.
- This modifies the resonant frequency of the ring:
Allows filtering a (tunable) specific wavelength from the optical signal.



$$\begin{aligned}\Delta n &\propto -\Delta N_e \\ \Delta \alpha &\propto \Delta N_e\end{aligned}$$

FREE CARRIER MODULATORS

3 main methods to modulate carrier density:

- Injection:
Apply forward-bias voltage to PiN diode structure in waveguide
- Depletion:
Reverse-biased PN diode to remove carriers from depletion region
- Accumulation:
MOS structure with thin layer of high-k gate oxide

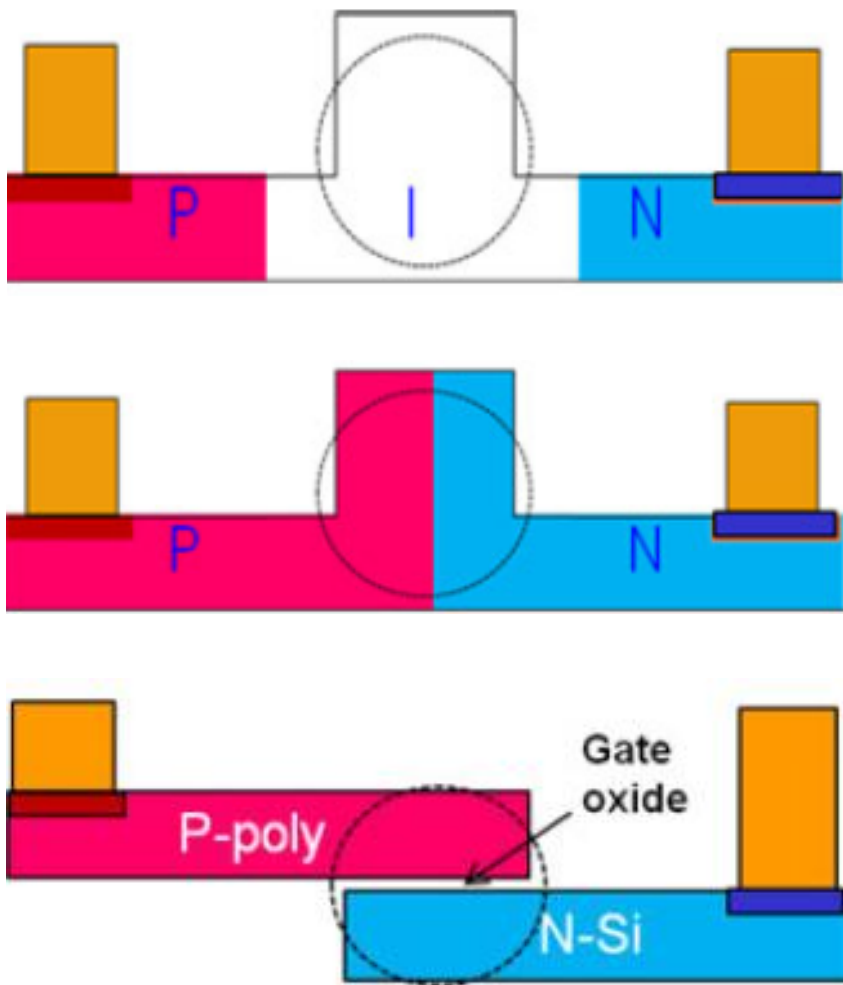


Figure of Merit	Carrier Accumulation	Carrier Injection	Carrier Depletion
Modulation Bandwidth	High +	Low -	High +
Extinction Ratio (ER)	Small -	Large +	Small -
Modulation Efficiency	High +	High +	Low -
Insertion Loss	High -	Low +	High -

SIMULATION OF CARRIER DEPLETION EFFECT

Free Charge density distribution with 4V reverse bias.

The depletion zone expands, as carriers are removed from the pn junction.

This changes the materials polarizability and response to electric fields.

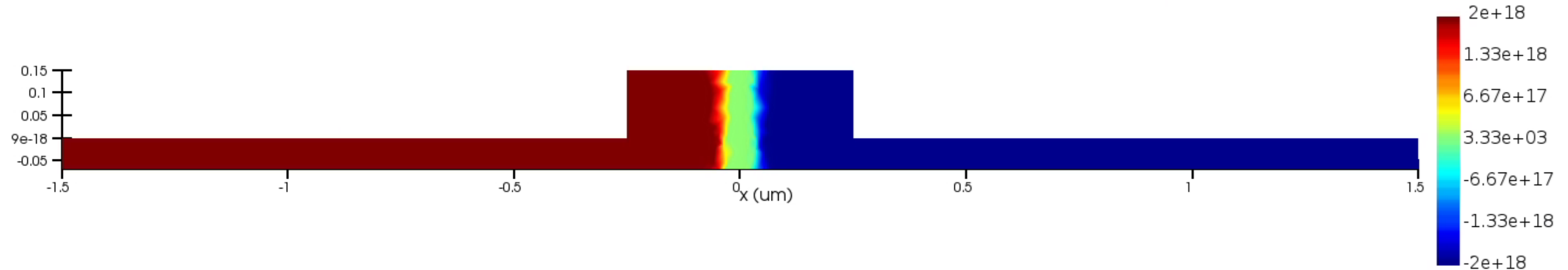
For Silicon at 1.55μm, the change in refractive index is given by Soref-Bennett model.

By removing charge carriers with reverse bias, the ΔN and ΔP are negative.

Results in increase of real refractive index, and a reduction in absorption coefficient.

$$\Delta n = - \left(\frac{e^2 \lambda^2}{8 \pi^2 c^2 \epsilon_0 n} \right) \left[\frac{\Delta N_e}{m_{ce}^*} + \frac{\Delta N_h}{m_{ch}^*} \right]$$

$$\Delta \alpha = \left(\frac{e^3 \lambda^2}{4 \pi^2 c^3 \epsilon_0 n} \right) \left[\frac{\Delta N_e}{m_{ce}^{*2} \mu_e} + \frac{\Delta N_h}{m_{ch}^{*2} \mu_h} \right]$$



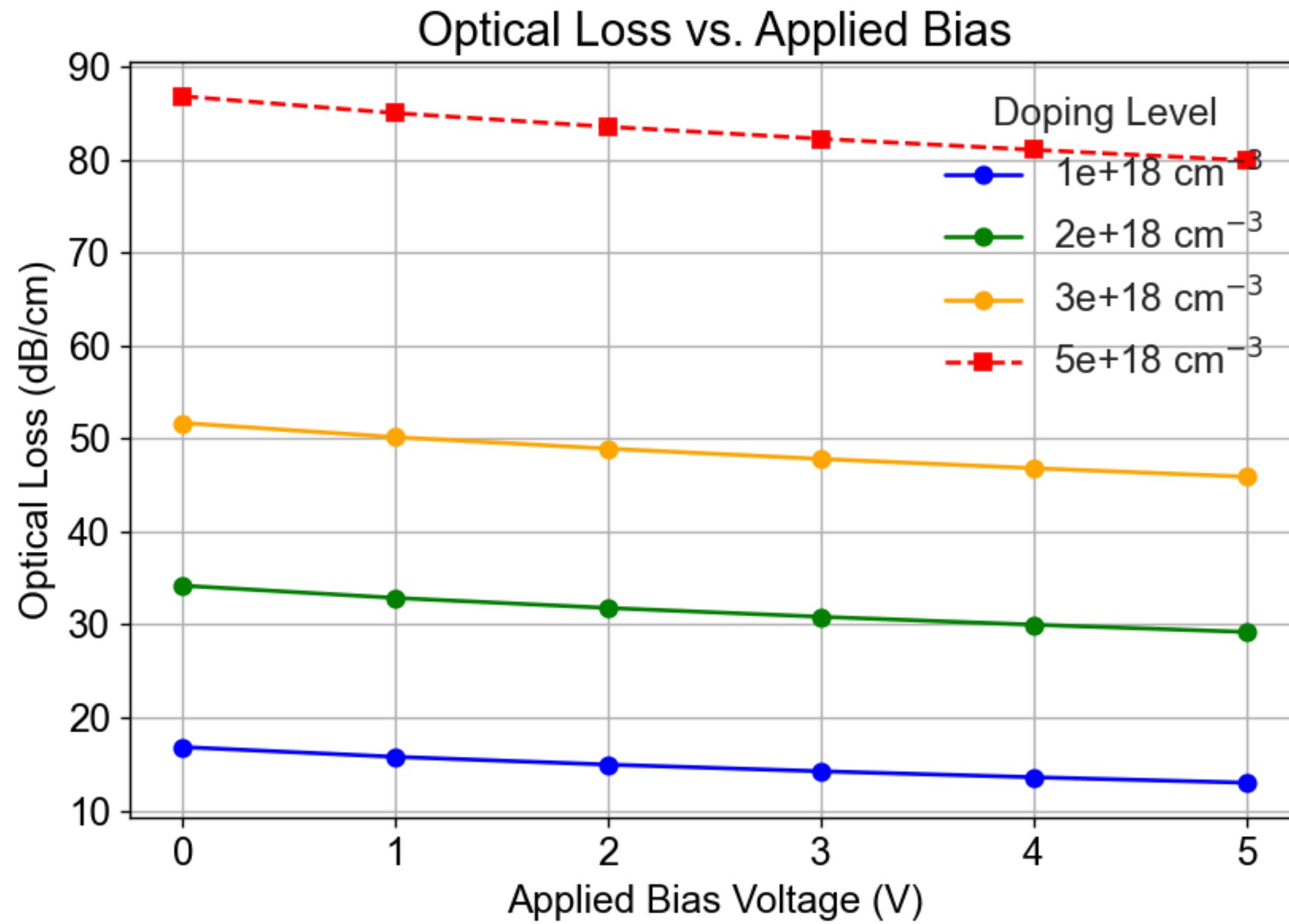
Soref and Bennett Model

Free-carrier plasma dispersion effect
@ 1.55 um

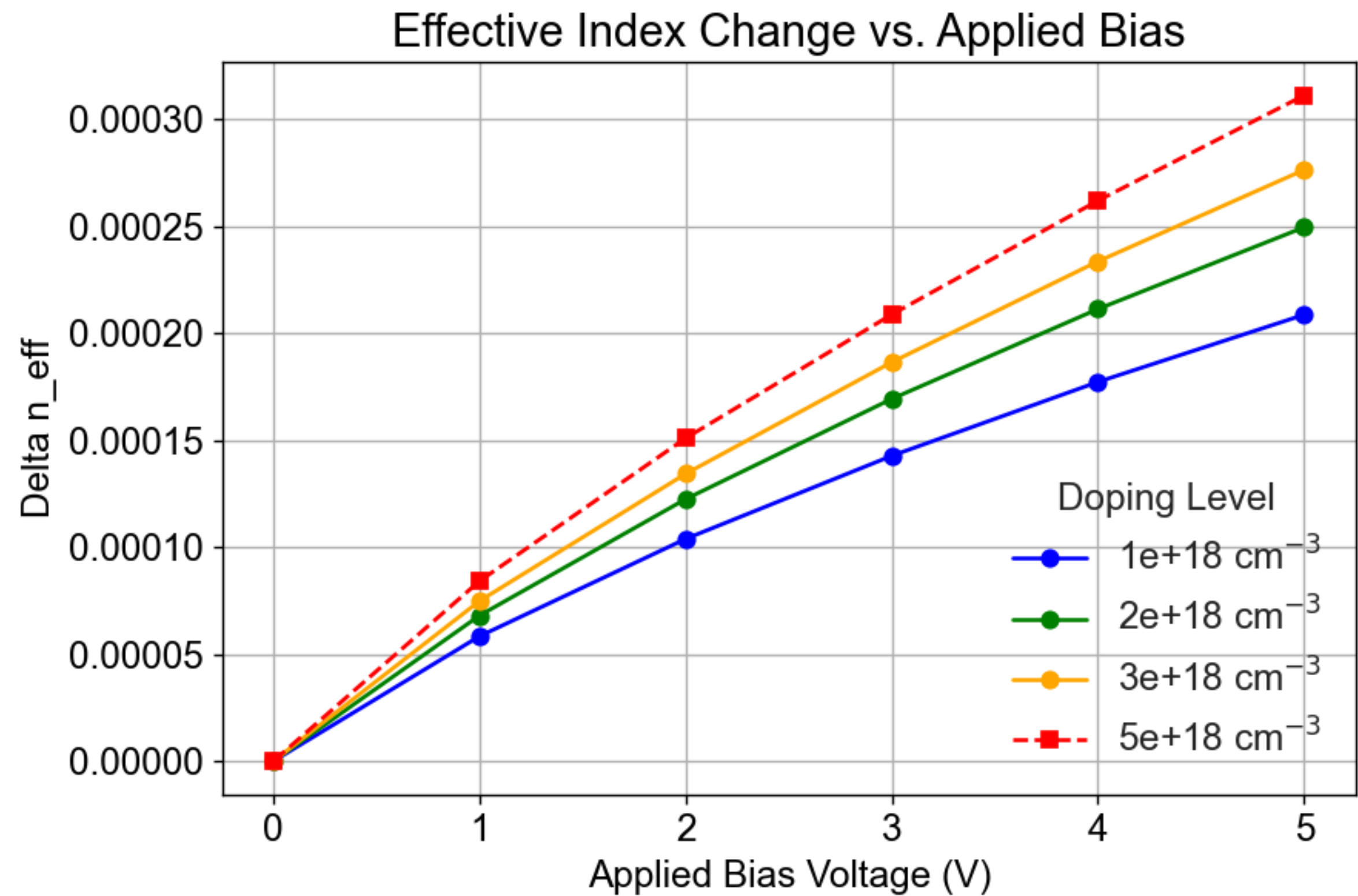
$$\Delta n = -8.8 \times 10^{-22} \times \Delta N - 8.5 \times 10^{-18} \times \Delta P$$

$$\Delta \alpha = 8.5 \times 10^{-18} \times \Delta N + 6 \times 10^{-18} \times \Delta P$$

LOSS



NEFF CHANGE



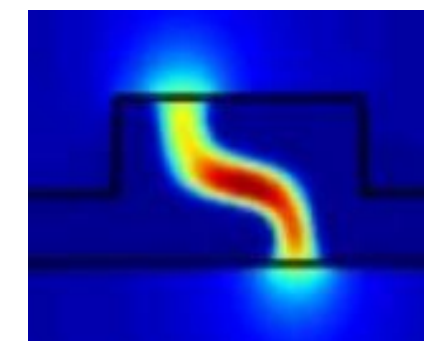
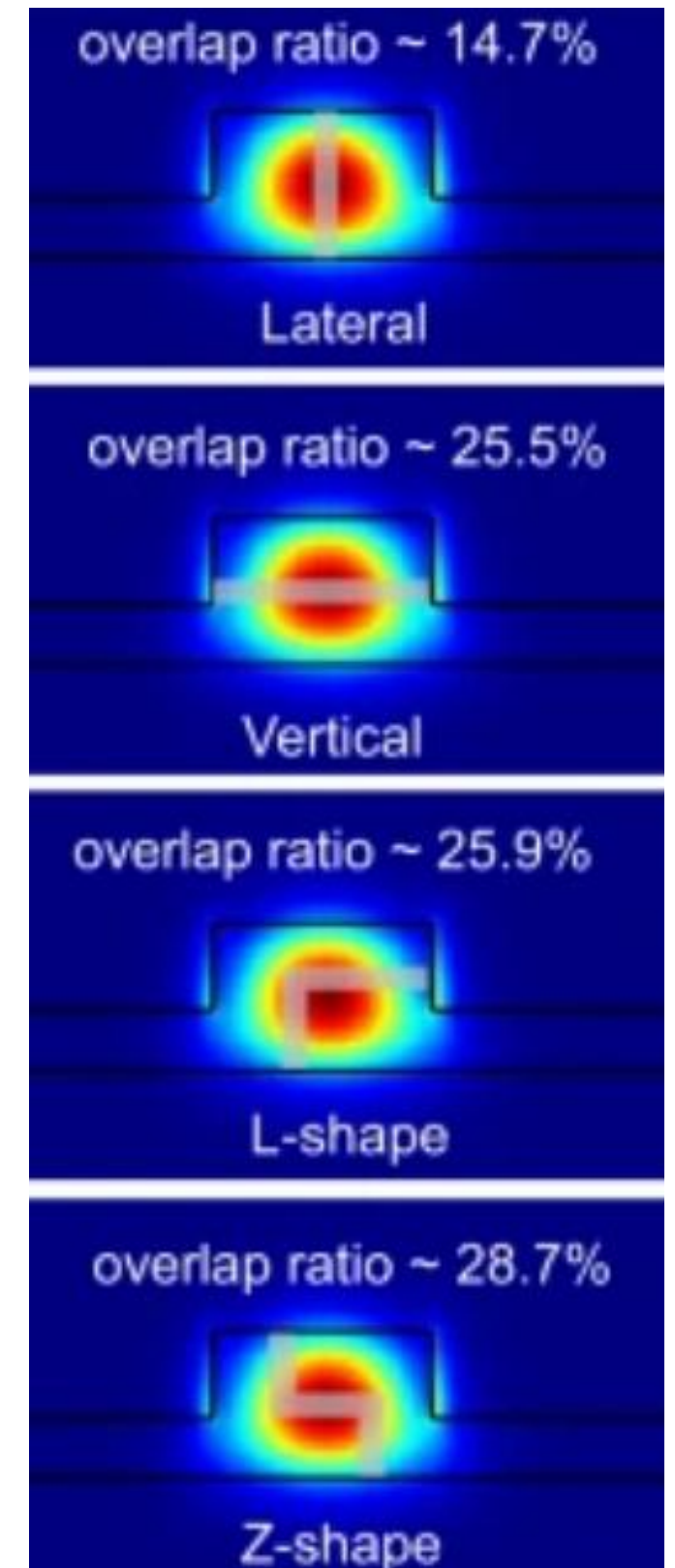
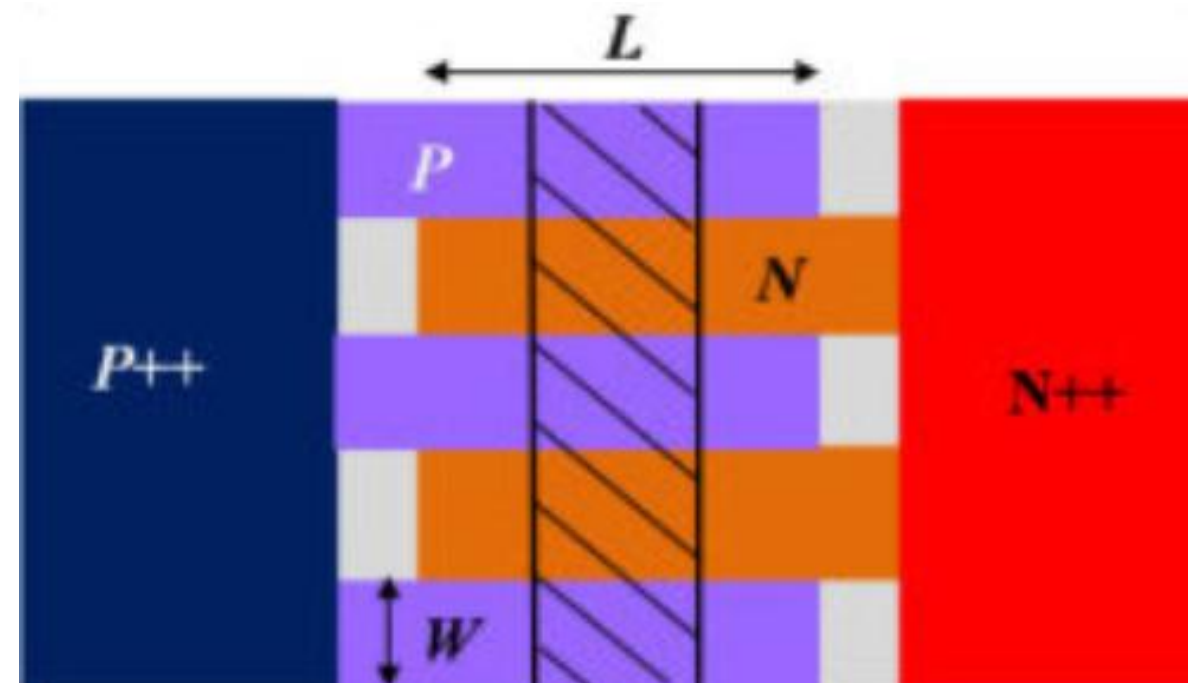
STATE-OF-THE-ART

Silicon Photonic Ring Modulator

- Doping: Z-shaped profile
- Compact size: radius $\sim 10\mu\text{m}$
- Bandwidth: $\sim 40\text{-}100\text{G}$
- Spectral bandwidth $\sim 20\text{nm}$
- Modulation Efficiency: $\sim 1.5\text{nm/V}$
- Propagation Loss $\sim 0.5\text{-}2\text{ dB/cm}$

Challenges:

- Co-packaged optics
- Thermal stability
- Integration of Laser sources



Active Components for 50 Gb/s NRZ-OOK Optical Interconnects in a Silicon Photonics Platform

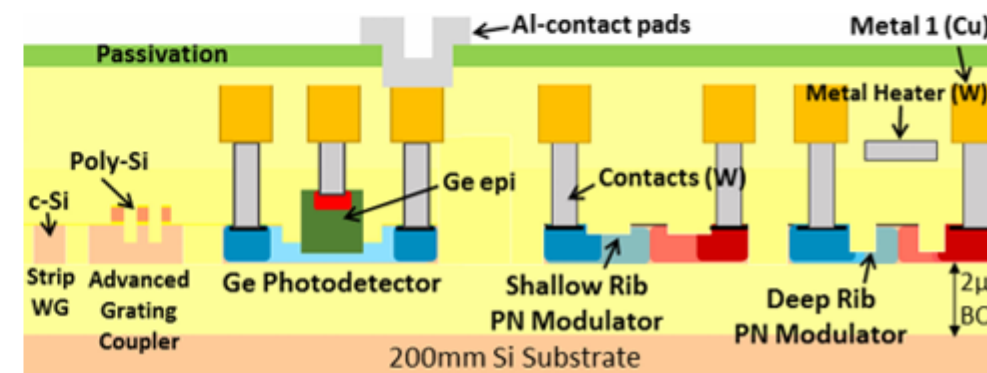
M. Pantouvaki, *Member, IEEE*, S. A. Srinivasan, Y. Ban, P. De Heyn, P. Verheyen, G. Lepage, H. Chen, J. De Coster, N. Golshani, S. Balakrishnan, P. Absil, and J. Van Campenhout, *Member, IEEE*

(Invited Paper)

Abstract—We present active components developed in imec's silicon photonics platform that enable 50-Gb/s non-return-to-zero operation using CMOS compatible voltages.

Index Terms—Electrooptic modulators, integrated optoelectronics, optical interconnects, optical receivers, optical transmitters, photodetectors, photonic integrated circuits, silicon photonics.

I. INTRODUCTION



SUMMARY OF PAPER

SUMMARY OF PAPER

Silicon Photonic Ring Modulator

Motivation:

Datacenters demand broader bandwidth and higher-density interconnects.

Presented:

High-performance Active (modulator, detector) & Passive components co-integrated with CMOS compatible process - imec iSiPP 200mm platform – for optical interconnects.

JOURNAL OF LIGHTWAVE TECHNOLOGY, VOL. 35, NO. 4, FEBRUARY 15, 2017

631

Active Components for 50 Gb/s NRZ-OOK Optical Interconnects in a Silicon Photonics Platform

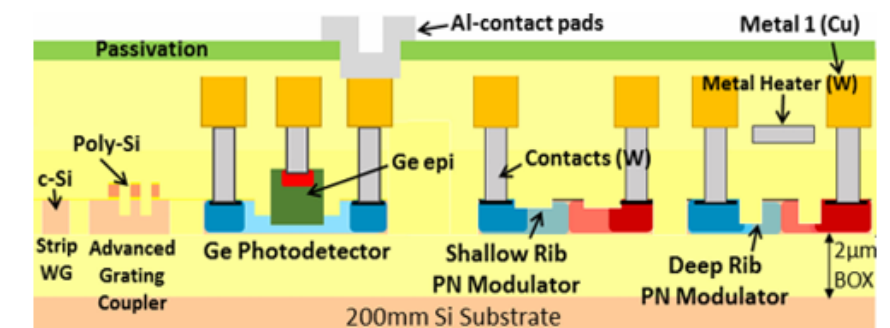
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I. INTRODUCTION



SUMMARY OF PAPER

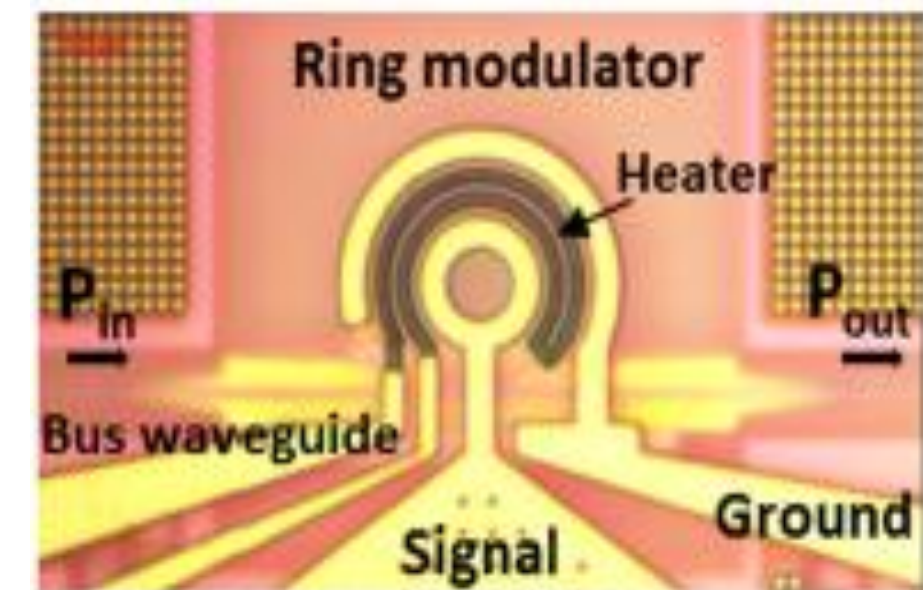
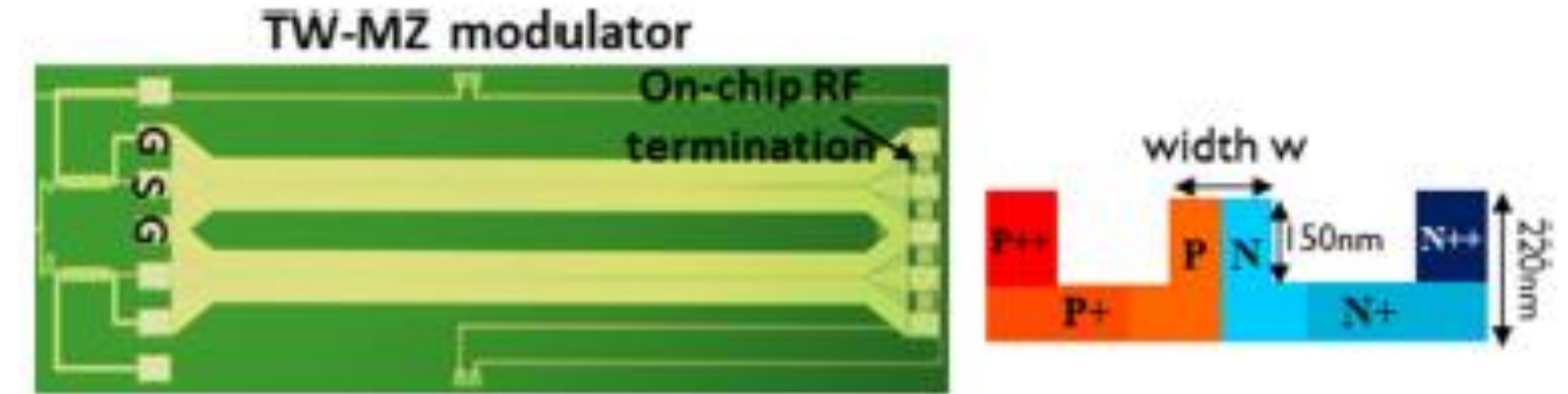
Silicon Photonic Ring Modulator

Modulator:

Comparison between TW-MZ, Ge-EAM and Ring modulator.

Ring modulator advantages

- Compact footprint (radius $5\mu\text{m}$)
- Low drive voltage (1-2 V_{pp})
- High speed (BW 47 GHz)
- Efficient (5-12 fJ/bit)
- Tunable, suited for dense WDM



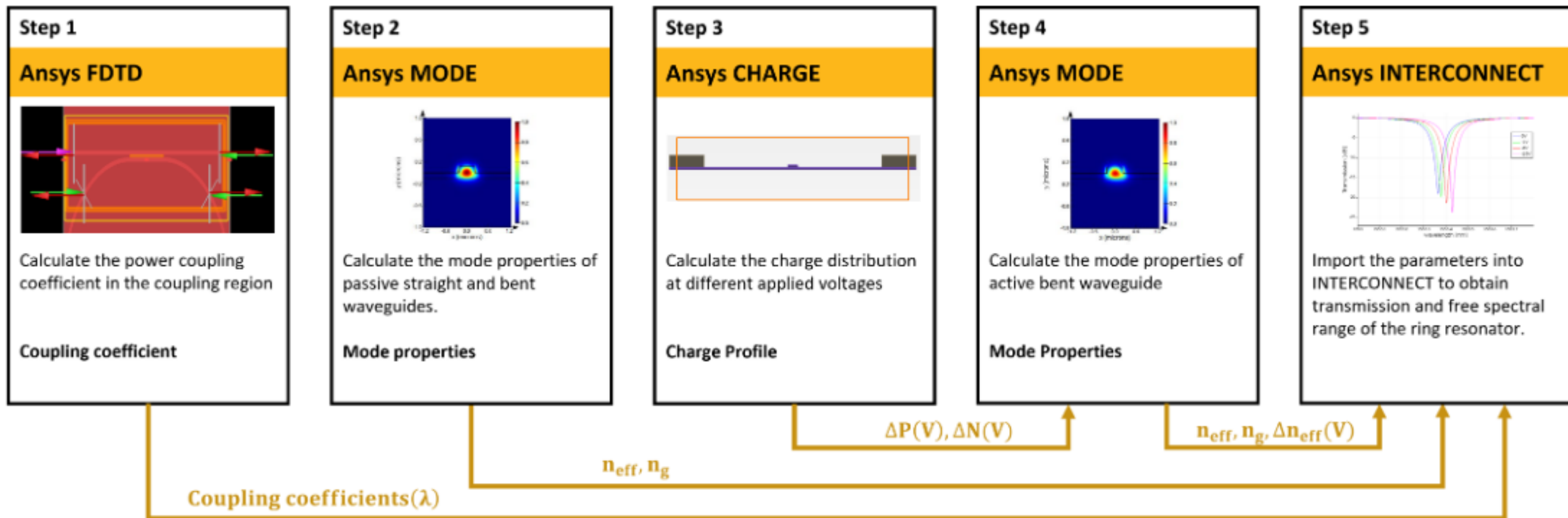
Main trade-off in ring modulator:

Bandwidth (speed) vs. Extinction Ratio (Wavelength selectivity).

PLAN FOR SIMULATION

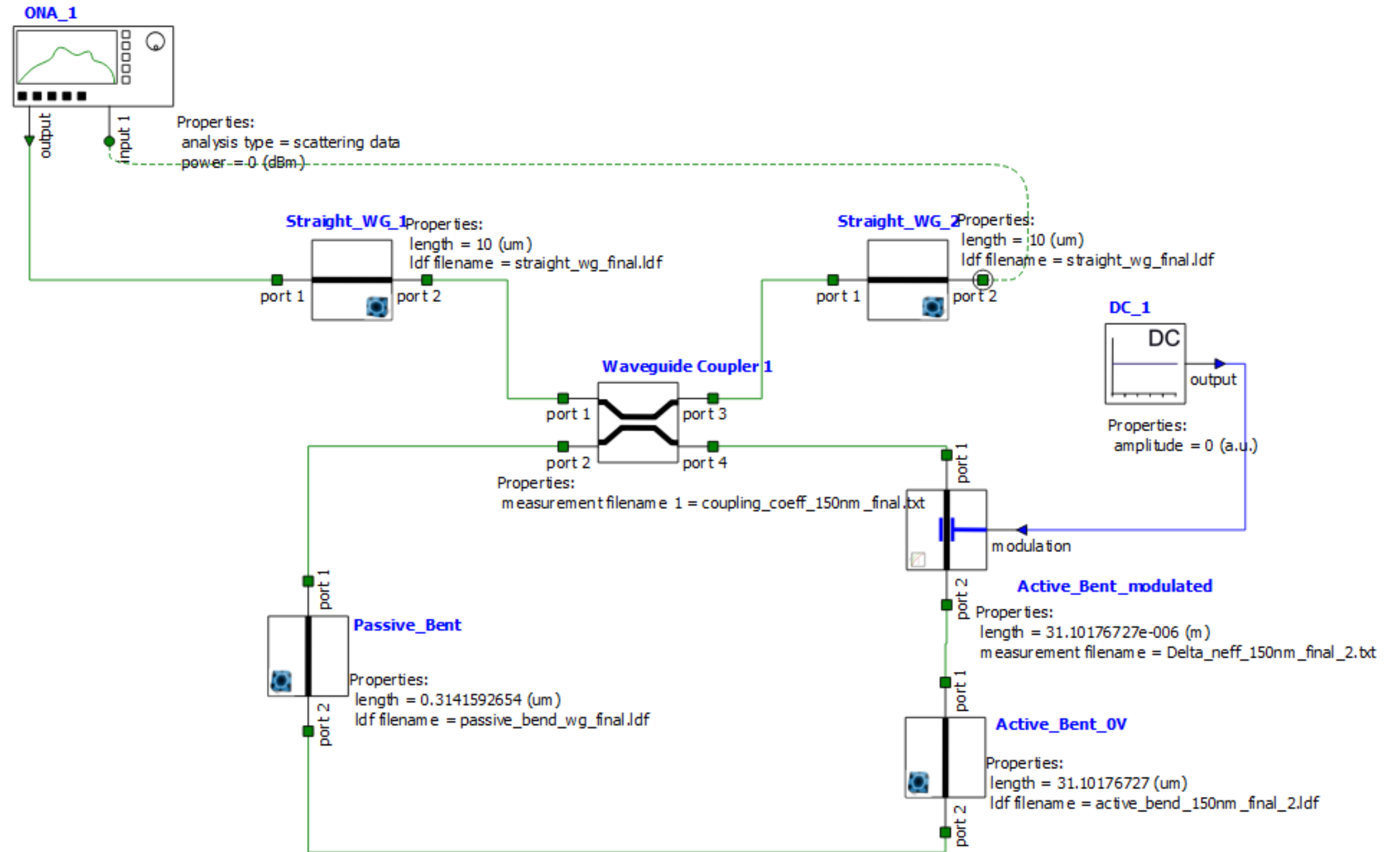
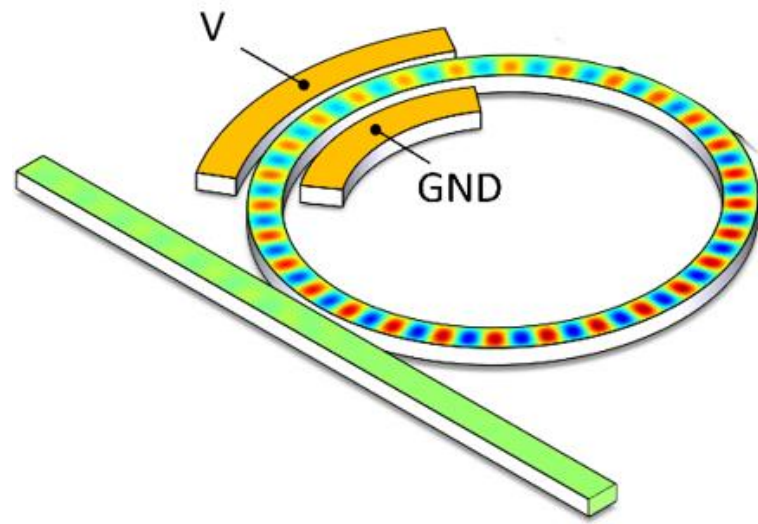
Overview

Understand the simulation workflow and key results



The essential building blocks of the ring and racetrack modulators are (1) a waveguide coupler, (2) passive waveguides, and (3) active waveguides.

LAYOUT → SCHEMATIC



RESULTS (1)

Doping: $5e18 \text{ cm}^{-3}$

Gap: 150 nm

Q-Factor: 3665

Modulation Efficiency: 33.992 pm/V

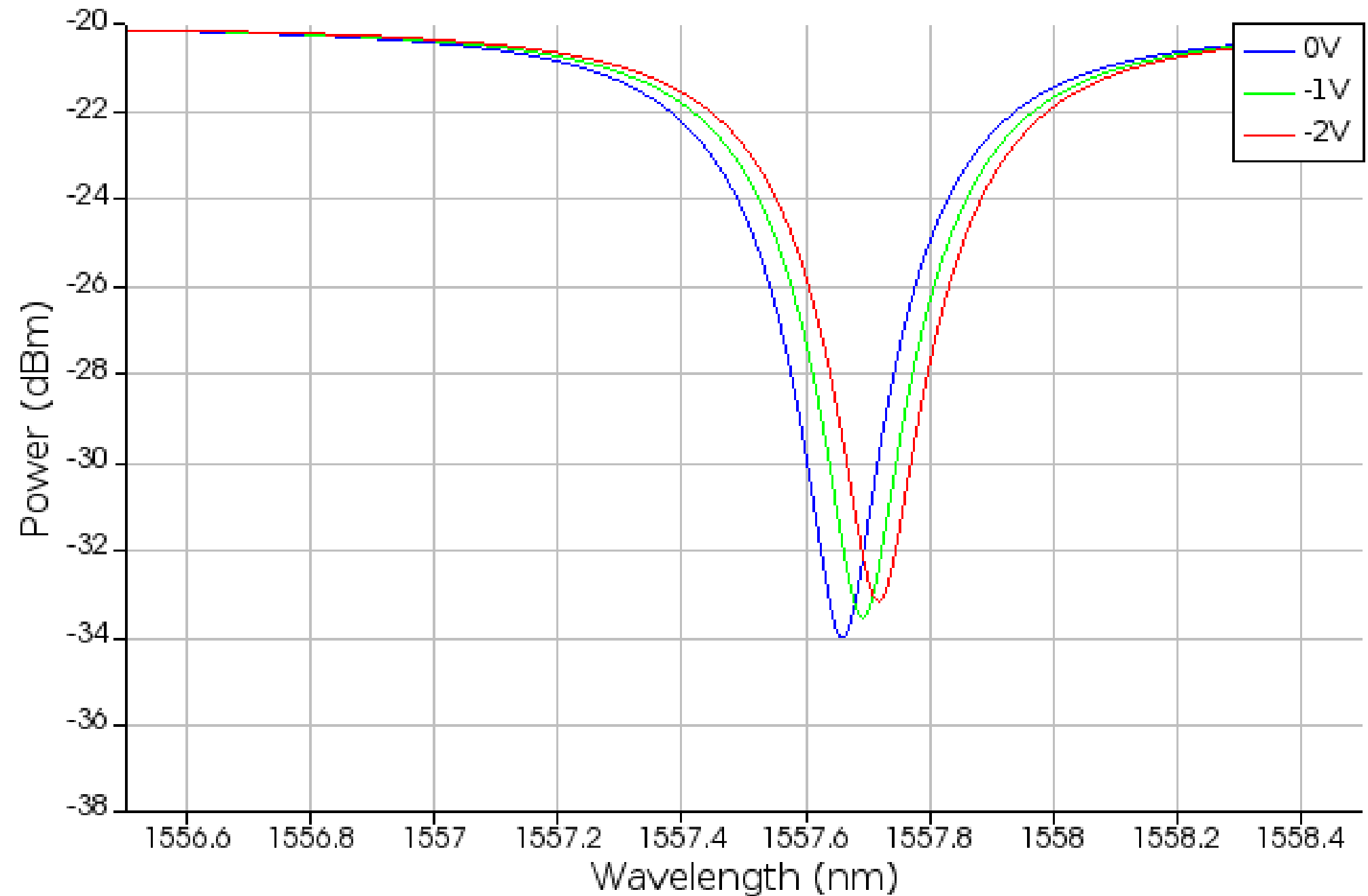
FSR: 19.104 nm

From Paper:

Q-Factor: 2200

Modulation Efficiency: 45.1 pm/V

FSR: 19.3 nm



RESULTS(2)

Doping: $2e18 \text{ cm}^{-3}$

Gap: 230 nm

Q-Factor: 9865

Modulation Efficiency: 26.73 pm/V

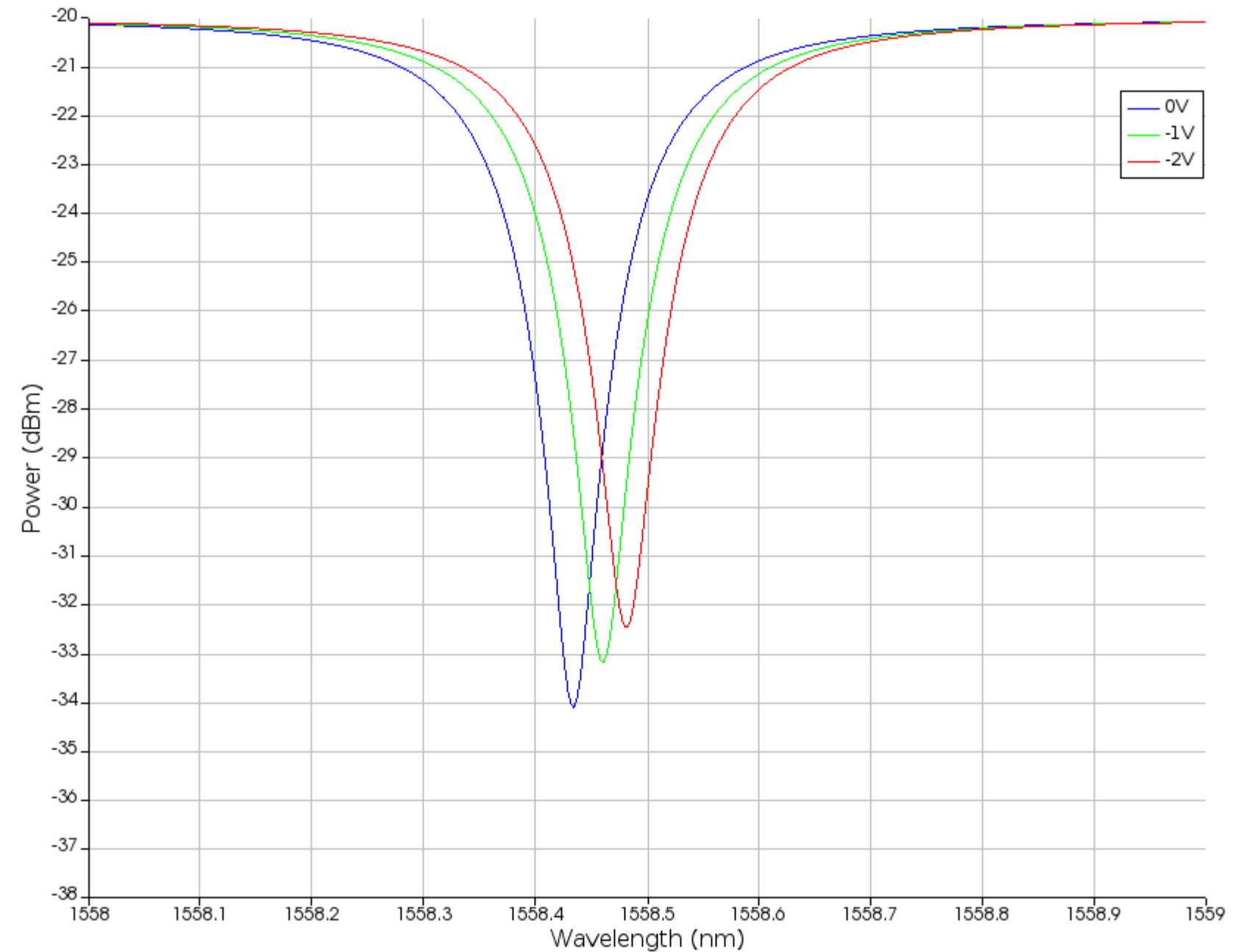
FSR: 19.113 nm

From Paper:

Q-Factor: 9900

Modulation Efficiency: 30.1 pm/V

FSR: 19.3 nm



MODULATION SPEED

- Reduced Q factor implies reduced photon lifetime. $\tau = \frac{Q\lambda}{2\pi c}$

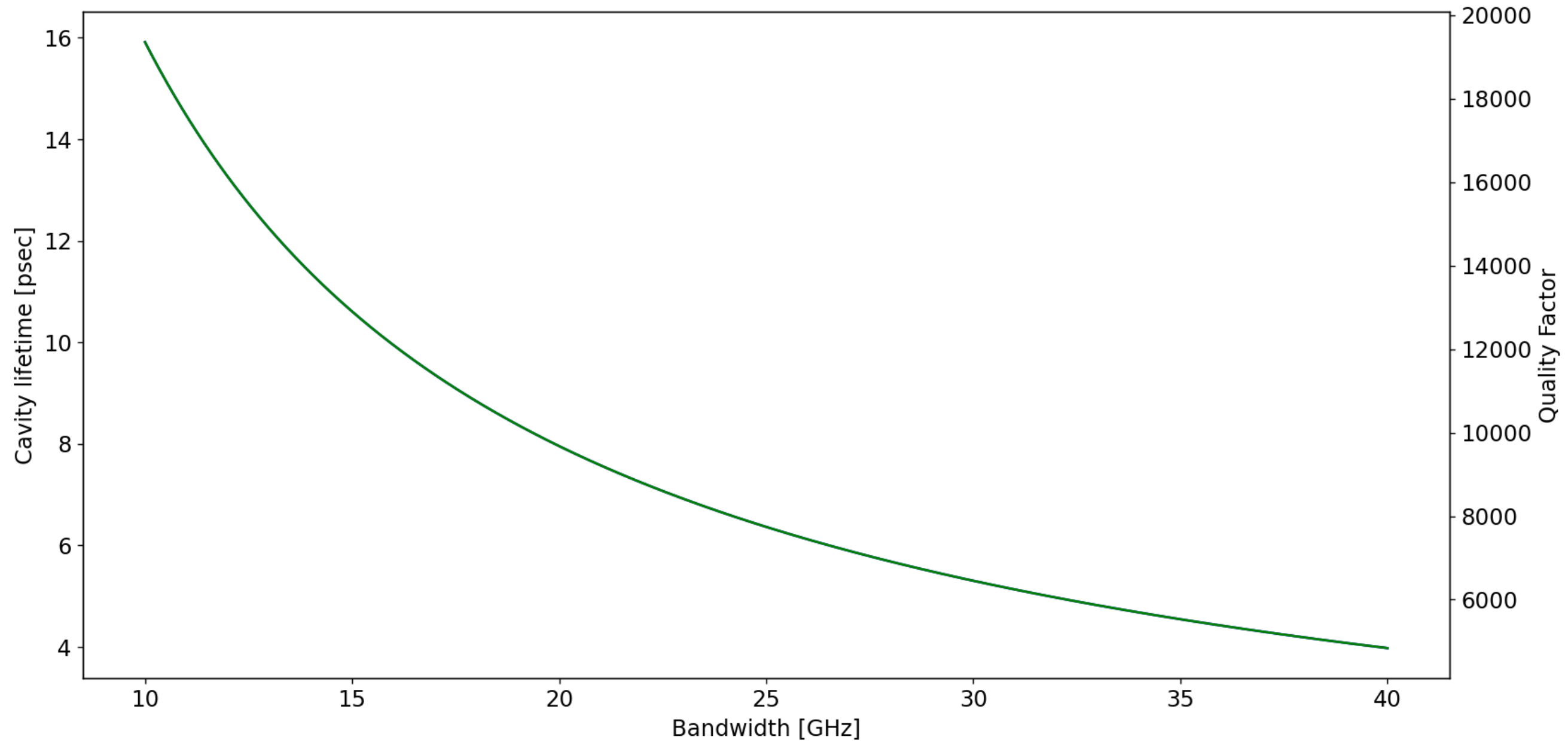
Open discussion for the class:

- Will the Q-factor impact the modulation bandwidth? **And why?**

$$\frac{1}{f_{3dB}^2} = \frac{1}{f_Q^2} + \frac{1}{f_{RC}^2} \qquad f_Q = \frac{1}{2\pi\tau} \qquad f_{RC} = \frac{1}{2\pi RC}$$

Q-Factor	f_Q	f_{RC}	f_{3dB}
9900	19.5 GHz	43 GHz	16 GHz
2200	88 GHz	51 GHz	47 GHz

PHOTON LIFETIME



PERFORMANCE AND FINAL RESULTS

- Parameters match well with the reference design.
- Small geometry/loss differences affect performance.
- Q-factor and ER follow expected trends.

TABLE 1 - DEVICE PARAMETERS & PERFORMANCE

Parameter	Paper (Pantouvaki et al., 2017)	Our Simulation
Ring radius R	5μm	5 μm
Waveguide width	500 nm	500 nm
Slab thickness	70 nm	70 nm
Coupling gap	150–230 nm sweep	150-230nm
Doping Conc.	2e18-5e18 (Lateral PN)	2e18-5e18 (Lateral PN)
Modulation voltage	1 Vpp	1 V DC tuning in Lumerical
Laser wavelength	1550 nm	1550 nm

TABLE 2 - OPTICAL CHARACTERISTICS

Optical Parameter	Paper	Our Simulation
Q-factor (high-Q)	~9900	~9865
Q-factor (low-Q)	~2200	~3665
ER (Extinction Ratio)	12–15 dB	~12-14 dB
Resonance shift (pm/V)	25–50 pm/V	~55 pm/V
FWHM	0.1 nm (high-Q), 0.22 nm (low-Q)	0.09 nm (high Q), 0.3 nm (low-Q)
FSR	~19.3 nm	~19.1 nm

TABLE 3 - SPEED PERFORMANCE

Frequency Parameter	Paper	Our Simulation
Photon lifetime limit f_ph	19 GHz (high-Q), 88 GHz (low-Q)	~19.5-52GHz

FUTURE POSSIBILITIES

- Optimize coupling gap for higher Q and ER
- Add dual-bus design for add-drop functionality
- Explore high-speed drivers for >50 Gb/s operation
- Integrate thermal tuning for wavelength stability
- Monolithic integration of Laser sources

REFERENCES

- Microsoft PowerPoint - lecture11 ee689 rrm tx
- A 5 × 200 Gbps microring modulator silicon chip empowered by two-segment Z-shape junctions | Nature Communications
- Ring Modulator – Ansys Optics
- Ring Resonator Modulators in Silicon for Interchip Photonic Links | IEEE Journals & Magazine | IEEE Xplore
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